Do Household Inflation Expectations Respond to Macroeconomic Data Releases?

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Abstract

This paper investigates how household inflation expectations respond to macroeconomic data releases. By utilizing an unexploited feature of the Michigan Survey of Consumers (MSC) – the interview dates of each respondent between June 1979 and May 2019 – I identify whether respondent expectations differ within tight windows around data releases. I consider 480 monthly releases of the Employment Situation Summary (ESS), Consumer Price Index (CPI), Industrial Production (IP), and Gross Domestic Product (GDP), as well as 459 Federal Open Market Committee (FOMC) meetings/calls. I find that household inflation expectations significantly respond to information about the unemployment rate and CPI inflation, suggesting salience of these variables for households relative to others considered. This includes the size and sign of the data release relative to the previous period data release. With 40 years of interviews, this paper also addresses whether the relevance of these releases differs during recessions. I find that information about the unemployment rate is particularly salient preceding, during, and following recessions. For a shorter subsample, this paper concludes with whether deviations in these releases from market expectations, as measured by the Bloomberg consensus forecasts, prompts adjustment. I find that household inflation expectations respond to deviations in IP and real GDP growth, indicating that unanticipated changes in output growth are relevant.

JEL Classifications: E31, E52, E71

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1 Introduction

Inflation expectations are the most important driver in actual inflation.
- Jerome Powell, Chair of the Federal Reserve, February 2019

The effectiveness of monetary policy hinges crucially on policymakers' ability to influence the inflation expectations of various economic agents in a variety of economic circumstances. Together with central bank mandates requiring the management of inflation realizations, a necessarily large weight is placed on understanding how inflation expectations are formed. The importance of such a connection between observed and expected inflation is not only documented within the academic literature[1], but clearly conveyed by the chief practitioners of monetary policy.[2] Indeed, this has motivated a burgeoning literature on understanding this formation process, particularly with respect to *what information matters* and *to whom the information matters*.

Theoretical and practical considerations suggest that such attention is warranted, with the former following from the standard implementation of the Fisher equation. This dictates that the real interest rate, a key determinant of agents' intertemporal decision-making, is the gap between the nominal interest rate and the expected inflation rate of agents. Since policymakers determine the path of the nominal interest rate, monetary policy actions are propagated via changes in the perceived real interest rate, the extent of which reflect subsequent changes in the expected inflation rate of agents. This establishes that, in theory, inflation expectations are first order to the effectiveness of monetary policy actions.

Practical implementation of monetary policy generally intends to operate via this same real interest rate channel. As indicated by the introductory quote, influencing the inflation expectations of agents is paramount to central bank strategies to change the perceived real interest rate, subsequently affecting economic activity. The past three decades of central bank conduct indicates as much, wherein the implementation of expectation-based strategies in meeting objectives and mandates has only increased in prevalence. This is exemplified by the wide adoption of explicitly stating intended policy changes, as well as providing forward-guidance about future policy actions. However, the reality of influencing the inflation expectations of agents is not without numerous challenges, chief amongst them being that the formation of

1: See recent work from Binder and Kamdar (2022) and IMF (2023).

2: Jerome Powell September 2021: *Inflation expectations are terribly important. We spend a lot of time watching them.*, and Janet Yellen October 2016: *... we need to know more about the manner in which inflation expectations are formed and how monetary policy influences them.*

inflation expectations is markedly different for different types of agents.[3] This reformulates the implicit objective of policymakers as needing to influence the different types of inflation expectations, instead of influencing inflation expectations more broadly. As such, a deeper understanding of the expectation formation process across different types of agents is required, above and beyond the role of monetary policy alone.

Of particular importance for this paper is that *household* inflation expectations are distinct from those of firms, professional forecasters, and market participants. In contrast to the other groups, household inflation expectations are characterized by significant cross-sectional dispersion and notable upward bias with respect to observed inflation rates, both almost certainly driven by a multitude of idiosyncratic factors. This presents an econometric challenge in identifying changes in household inflation expectations that are attributable to specific events like monetary policy announcements. Additionally, if household inflation expectations largely reflect localized economic signals, then there is an apparent disconnect between their relevant information and the relevant information for policymakers. Put differently, since monetary policy unambiguously reflects contemporaneous aggregate economic information, is this information also relevant in the formation of household inflation expectations?

Utilizing unexploited data in the Michigan Survey of Consumers (MSC), this paper focuses on whether household inflation expectations respond to such information, namely macroeconomic data releases. Conducted and released monthly since January 1978, the MSC is the longest running survey of U.S. households, and elicits both qualitative and quantitative expectations. Despite its extensive use in various branches of macroeconomic research, this is the first paper to document that, beginning in June 1979, the interview date of *each* respondent has been recorded and is publicly available. Previously only the survey release that each respondent was interviewed for was known, presenting a major identification problem. For example, while responses from the 602 interviews conducted for the May 2018 survey are available, no method of discriminating between the different information sets of the respondents existed. The lack of such a method, in turn, prevented attributing changes in household inflation expectations to particular macroeconomic releases, since the releases confound each other.

With the interview dates of the more than 250,000 MSC respondents between June 1979 and May 2019, this paper circumvents the previous type of identification problem by identifying interviews occurring within one to three days of five types of macroeconomic releases. These include 480 monthly releases of the Employment Situation Summary (*ESS*), Consumer Price

3: While not exhaustive, see Candia et al. (2023), Cavallo et al. (2014), Coibion et al. (2018a), Coibion et al. (2018b), D'Acunto et al. (2022), Ehrmann et al. (2015), Falck et al. (2021), Mankiw et al. (2003), and Weber et al. (2022).

Index (*CPI*), Industrial Production (*IP*), and Gross Domestic Product (*GDP*), as well as 459 Federal Open Market Committee (*FOMC*) meetings and calls.[4] This quasi-experimental setting thereby allows for estimation of how household inflation expectations respond to each of the considered releases, without confounding the effect due to all of the other releases. Of course, releases have historically occurred within close proximity to each other, but because they have not *always* done so, this confounding effect can be controlled for.[5]

For this 40 year period, I explore whether household inflation expectations respond to two characteristics of the information conveyed within each type of release. Throughout this paper, I refer to these as the *sign* and *size* of the release, wherein each is referring to the relationship between the new estimates of the releases headline series with respect to its previous estimates. As an example, if the most recent release of the *ESS* suggests the unemployment rate was 5 percent, while the preceding release suggested is was 4 percent, then the *sign* of the release would be positive and the *size* would be 1 percentage point (p.p.). While the economic interest of expectations adjusting to the *size* of changes is evident, the interest in the information conveyed by the *sign* of the change may ambiguous. To alleviate this, one may simply consider the possibility that households make no economic distinction between the inflation rate being 0.1 p.p. or 0.2 p.p. higher than was last known, instead only incorporating the fact that both indicate a *higher* inflation rate. Collectively, I find that household inflation expectations do respond to macroeconomic data releases, with the relevant information from these releases being the unemployment rate and CPI inflation. Notably, this coincides with the dual mandate of the Federal Reserve, highlighting the potential for the central bank to, in part, manage household inflation expectations via standard policy actions consistent with its mandate.

First, I find that household inflation expectations respond strongly to the *size* of changes in the unemployment rate, where a 1 p.p. *increase* causes household inflation expectations to *decrease* by about 0.6 p.p. Accounting for the possibility that households asymmetrically adjust in light of increases or decreases, I find that expectations respond more to *increases* in the unemployment rate than to decreases. Similar to the first estimates, these suggest that a 1 p.p. *increase* in the unemployment rate prompts a *decrease* in household inflation expectations of about 0.7 p.p. In contrast, I find that household inflation expectations respond to the *sign* of changes in headline CPI inflation, but only when the headline rate is lower than was previously published, regardless of how much lower it is. Further, I do not find that the *size*

4: The *ESS* includes series like the unemployment rate and the number of jobs added. For all the series, I utilize the real-time values as they appeared at the time of the release. For GDP, the monthly aspect refers to the initial estimate, released during the first month of the following quarter, and both subsequent revisions.

5: Additional details are in Section 3.

of these changes prompts adjustment, nor do households adjust expectations more in response to increases or decreases.

Next, I consider whether the relevance of these releases change over the business cycle, an understanding of which is of particular importance for policymakers. For primarily two reasons, this question has not been addressed in the literature employing high frequency identification strategies. First, similarly aligned literature has relied on either conducting single instance surveys or conducting multiple surveys over short periods of time. In such a setting, the sample restrictions present would prohibit identification of how household inflation expectations respond to releases occurring outside recessions, compared to those occurring during recessions. Second, utilizing more micro-econometric identification strategies in addressing macroeconomic questions generally gained traction in the wake of the financial crisis. This led to the establishment of some new surveys of household inflation expectations where the same identification strategy has been employed, but these new surveys simply haven't been conducted long enough to consider such a question.

Using the historical dates of NBER recessions, I estimate separate responses of household inflation expectations to releases occurring outside of recessions and those occurring during recessions. Consistent with my previous findings, household inflation expectations respond strongly to information about the unemployment rate, both outside and during recessions. For releases occurring outside recessions, household inflation expectations *decrease* by about 0.5 - 0.7 p.p. for a 1 p.p. *increase* in the unemployment rate. While identification of the response to the *sign* of releases during recessions is generally not possible, I do find that household inflation expectations are about 0.2 - 0.3 p.p. lower after recession releases of the *ESS*. Since nearly all of the recession estimates of the unemployment rate would indicate that it is increasing, one may conclude that households are lowering their inflation expectations in response to the increasing unemployment rate.

Due to sheer practicality, NBER must assert official start and end dates to recessions. However, it is not immediately clear why, for example, the December 2007 release of the Consumer Price Index could not be considered as having occurred in a recessionary-like state of the economy, considering the next month is the official start of the 2008 financial crisis. Explicitly, the adjustment of household inflation expectations may differ both preceding and following official recessions, as well as during recessions. To assess this, I additionally allow for asymmetric responses to releases in the six months preceding and following recessions. I find that household inflation expectations respond strongly to the size of changes in both the unemployment rate and CPI inflation in the 6 months following recessions.

Finally, it could well be argued that the response to changes in the headline series is less important than how household inflation expectations respond when these releases deviate from expectations. This can viewed as decomposing whether household inflation expectations respond to anticipated or unanticipated changes in these releases, where the latter is typically denoted as the *surprise*. To measure the unanticipated changes, I incorporate the consensus forecast from Bloomberg for each release occurring between November 1996 and May 2019.[6] Instead of considering the *sign* and *size* of changes in the releases with respect to their previous values, I instead classify these as relative to the consensus forecast, so whether the value was higher or lower than expected, and by how much, respectively. I find that household inflation expectations do not respond to the *sign* of any of the surprises. However, the estimates suggest that what matters is the *size* of the surprises in IP and real GDP Growth, in both cases suggesting that *higher* than expected growth rates *decrease* household inflation expectations.

The paper proceeds as follows. Section 2 discusses related literature on household inflation expectations, focusing particularly on those employing high-frequency identification strategies. Section 3 provides background on the MSC, details the availability of the interview dates, and further describes this papers identification strategy. Section 4 presents this papers baseline results, Section 5 presents results for recessions, and Section 6 presents results when incorporating the expectations of forecasters. Section 7 summarizes the key findings of this paper and concludes.

2 Related Literature

While impossible to exhaustively discuss all of the relevant literature on household inflation expectations, work most closely related to this paper generally employs a high frequency identification approach to assess how households adjust expectations around macroeconomic events.[7] Amongst these, the closet work appears in Binder et al. (2022). By using the interview dates of participants in the Federal Reserve Bank of New York's Survey of Consumer Expectations (FRBNY SCE), they identify how household inflation expectations respond to a series of macroeconomic events. Such events include FOMC announcements, macroeconomic data releases, as well as news about the Covid-19 pandemic. They find these events affect household inflation expectations in different ways, i.e. households do not always adjust expectations

6: Work on how *professional* forecasters respond to macroeconomic information include Kliesen and Schmid (2004), Bauer (2014) and De Haan et al. (2020).

7: Forgoing a more general discussion of the information treatment approaches, see Agarwal et al. (2022), Armantier et al. (2016), Armantier et al. (2022),

in the same way to similar events. For example, their findings suggest that FOMC meetings with target rate cuts sometimes prompt a large increase in inflation expectations, whilst also sometimes prompting a large decrease.

Although their work overlaps notably with this paper, a brief detailing of the differences between the FRNBY SCE data they use and the MSC data used in this paper is needed before the comparison of results. This is because differences are primarily derived from differences in each of the surveys, more or less with each having benefits and limitations. First, the FRBNY SCE interviews around 1300 households each month, nearly triple the 500 interviewed for the MSC. This amounts unambiguously to a benefit of using the SCE data, as the density of interviews around specific *instances* of a particular macroeconomic event is much larger than it is in the MSC data. Specifically, this allows them to identify *separate* effects for each release occurring for the duration of the survey, whereas the number of interviews conducted for the MSC would typically not support such an identification claim. Thus, the MSC data requires some amount of pooling across separate releases of the same type to identify an average response on the part of household inflation expectations.[8] Although, notable is the observation that, between June 2013 and May 2019, the sub-sample for which both papers share, more than 60 percent of CPI and IP releases occurred within 2 days of each other, which is the window that their work uses.

The second difference is simply the duration of which each survey has been conducted. The MSC has been conducted in various forms since the end of World War 2, whereas the FRBNY SCE conducted its first official survey in June 2013. This limits consideration of how household inflation expectations account for macroeconomic releases to mostly the ZLB and Covid-19 pandemics, and does not present an opportunity to understand the relevance of such information in more standard parts of the business cycle. This difference in these durations additionally manifests as different focuses on what information from each release is considered.

As briefly discussed in the introduction, this work focuses, by in large, on characteristics of changes in headline series. For example, whether or not household inflation expectations adjust when the unemployment rate increases. In contrast, their work focuses on this adjustment process in respond to *surprises* in the releases, meaning the degree to which the headline series differs from what was expected by market participants. Thus, the most comparable results in this paper appear in Section 6, and should be compared against their pooled estimates. Using the same Bloomberg consensus forecasts that they employ but over a longer period, my results are inconsistent with their findings. Overall, I find that households respond to surprises in IP

8: This is discussed more extensively in Section 3. Broadly, the average number of interviews conducted around any particular release suggests that a minimum of about six instances need to be pooled for statistical purposes.

and real GDP growth, whereas they find significant responses to CPI releases.[9]

Binder (2021) conducts an online survey asking consumers whether they had recently heard news about inflation, and elicited their inflation expectations, around the June 10, 2021 release of the Consumer Price Index (CPI), which suggested the largest core inflation reading in nearly 30 years at the time. Given the narrow time frame around this release when consumers were interviewed, this provided quasi-experimental setting in which to interpret the casual effect of the CPI release on consumers inflation expectations. Notably, and discussed further momentarily, they find that consumers interviewed after the release were 11 percentage points more likely to have reported hearing news about inflation, but did not significantly adjust their inflation expectations in response. However, they did find that Inflation expectations only of highly-numerate respondents rose with the CPI release.[10]

Focusing on household inflation expectations around 12 FOMC press conferences, Lamla and Vinogradov (2019) conducts a series of interviews with consumers between December 2015 and June 2018. The proximity of the interviews in relation to the press conferences allows them to identify how household inflation and interest rate expectations respond to central bank communications. They find a strikingly similar result, wherein households interviewed after these FOMC press conferences were more likely to have reported hearing news about monetary policy without any statistically significant effect on households actual expectations.

Both Binder (2021) and Lamla and Vinogradov (2019) find the interesting result that households are more likely to report hearing news of these events, but that their inflation expectations are not significantly adjusted. As I discuss extensively in Section 3, I include respondent-level sentiment and news responses from the survey. If the releases considered do indeed increase the likelihood of a particular response to one of these questions, then their incorporation as control variables is problematic. However, assessing this suggests that, at least within the MSC, these releases do *not* increase the likelihood of respondents having heard news about a particular event. Further, I argue that this contradiction to their findings stems, more likely than not, from differences in the way the various surveys elicit information about the news a respondent has heard. In both of their surveys, because the intent was to assess the response of household expectations to specific types of events, ex-ante the news related questions were designed to ask specifically about each event. The design of the MSC news questions are

9: To be abundantly clear, neither paper attempts nor claims to exhaustively incorporate the possible macroeconomic releases that convey information to households. I am simply pointing out that differences in comparable results are almost surely driven in part by differences in the releases considered and the duration of available interviews.

10: News related literature: Hong et al. (2022), Shapiro et al. (2020) and Wang et al. (2020).

unambiguously not tailored to such considerations. Instead, the MSC questions aim to capture more general news-related topics that households may find important.

Finally, to cover the range of daily household surveys, Lewis et al. (2020) also exploits high frequency consumer interviews, wherein they use daily survey data from Gallup polls to gauge how monetary policy shocks affect households beliefs and expectations about economic conditions. Most notably, they document that households respond very quickly to certain types of information, including news about the federal funds target rate. This daily gallup data is also used in Singh and Mitra (2022). While the approach is different, they find that household inflation expectations respond strongly to movements in the unemployment rate, and that this response is stronger in light of negative information. This is strikingly consistent with what I find in both Section 4 and Section 5.

3 Data and Identification

As stated in Section 1, utilizing 40 years of interview dates for the respondents in the Michigan Survey of Consumers (MSC) serves as the foundation of the identification strategy employed in this paper. To this end, the first section provides some necessary background on the MSC, as well as detailing how the interview dates are publicly available. Subsequent sections focus on discussing identification and detailing additional data used throughout the remainder of the paper.[11]

3.1 Details for MSC and Interview Dates

Starting with some background on the MSC, the survey has been conducted in various forms since 1946, amounting to the longest survey of U.S. households elicting macroeconomic expectations. Beginning in January 1978, the survey has been conducted and released monthly, with survey administrators interviewing about 500 individuals, the composition of which is representative of the United States population.[12] Participation is determined via a random sample of telephone numbers in the coterminous United States, with respondents being asked about 50 questions ranging from qualitative questions about changes in their current economic situation and expectations, to quantitative point estimates about their inflation expectations.[13]

11: In the interest of brevity, this section intentionally omits a number of details that may be of interest to readers. Additional information about the survey, as well as some summaries, are provided in Section Appendix B.

12: see Section Appendix B.

13: A copy of the questionnaire used by the interviews can be found here. Additional information about the sample design of the survey can be found here.

For each survey, about $\frac{2}{3}$ of the respondents are new to the survey, with the remaining portion being respondents who were randomly selected to be interviewed again from the interviews conducted for the survey six months ago.

For the purposes of this paper, the important question(s) in the survey are those that elicit the respondents one year-ahead expected inflation rate, which contains two parts. First, the interviewer asks whether the individual believes that prices during the next 12 months will in general *go up*, *go down*, or *stay the same*.[14] In the case that respondents choose *stay the same*, the interviewer then follows up to clarify whether they mean that prices will go up at the same rate or whether they mean that prices will stay at the same level, as the former would suggest that the inflation rate will be the same whereas the latter suggests that the inflation rate will be zero. This yields the individuals' *qualitative* expectation about the sign of inflation over the next 12 months. The second part of this question, and the response that will be the dependent variable throughout this analysis, is the interviewer asking by what percent the respondents believes that prices will change by over the next 12 months, thus yielding the individuals point estimate.[15]

For each of the more than 250,000 respondents that have been interviewed since the monthly release of the survey began, the survey website makes the entirety of the micro-data available for download. For research utilizing the respondent-level responses, this is the standard way of acquiring the data. However, the survey makes no mention that this available micro-data is *not* all encompassing, wherein the survey administrators record additional information from each of the interviews that is not subsequently posted for direct download via their website. In other words, the micro-data that is made available on the surveys website is a *subset* of all the information that is collected from respondents at the time of their interview. Instead, this additional information *is* publicly available through the Inter-university Consortium for Political and Social Research (ICPSR), which is a social science repository that is also operated by the University of Michigan. Going forward, I will refer to these two different storage locations of the individual-level responses as the ICPSR data and MSC data.[16]

Effectively, one may think of the ICPSR data as the "raw" data collected throughout the process of interviewing respondents for a particular survey release. This data is then subsequently

14: The full text of the question is provided in Section Appendix B.

15: Additionally, if the respondent answers with a inflation rate of greater than 5% in magnitude, the interviewer is tasked with following up by asking *"Let me make sure that I have that correct. You said that you expect prices to go up/down by X percent. Is that correct?*. I refer the reader to D'Acunto et al. (2022) for discussion of this issue.

16: To be clear, the survey does explicitly mention that data from the surveys are stored on ICPSR, but is not made apparent that the data on ICPSR is more expansive nor that additional non-privileged information is available.

"cleaned", and then posted for download via the surveys website, i.e. the MSC data. To be clear, all of the data that is available within the MSC data is contained within the ICPSR data, but the converse is not true. One such piece of information collected that is not included in the MSC data, the piece of information that this paper utilizes for identification, is the interview date of each respondent. Within the ICPSR data, the day, month, and year that the interview was conducted is recorded, whereas the MSC data only contains the month and year for the survey that the respondents responses appear in. For example, an individual interviewed on March 29, 1994 would appear in the April 1994 survey results, with no mention in the MSC data that the interview was actually conducted in the previous month, but after the contemporaneous release of survey results for March 1994.

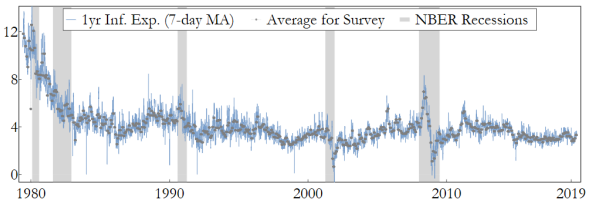
Although the survey began being conducted monthly in January 1978, recording the interview dates did not start until June 1979, where instead interviewers recorded the week of the month that the interviews were conducted. Additionally, because there is noticeable delay in uploading the necessary files to ICPSR, the last survey for which the interview dates are available is May 2019. Further, because the MSC data is a subset of the ICPSR data, both sources contain the same unique identifiers for each respondent, thus I augment the MSC data with these interview dates. In other words, while I was able to confirm the consistency of the ICPSR data with the MSC data, the only piece of additional information I utilize from the ICPSR data are the interview dates of the respondents, with all additional information coming directly from the MSC data.

As this is the first work to utilize daily frequency household inflation expectations from the MSC, discussion of some practical considerations of the series is necessary. To start, Figure 1 provides a historical perspective on the series along a number of margins. Within each panel of the figure, the horizontal axis spans the range of interview dates corresponding to individuals interviewed for survey releases between June 1979 – May 2019, with dates shown at the daily frequency. The first interview conducted for these surveys occurred on May 29, 1979, and the last interview was conducted on May 28, 2019. In all panels, the vertical axis is percent and the blue (solid) line is the seven day moving average of households' one year-ahead expected inflation rate, inclusive of the horizontal axis day.

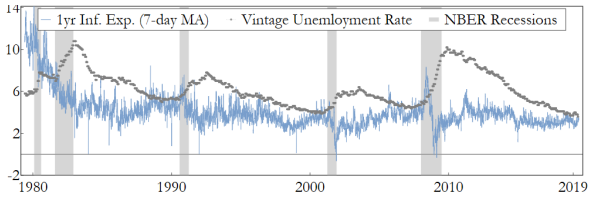
In panel (A) of the figure, the gray-connected dots are the average expected inflation rate for the survey that released that month, aligned so that the dot corresponds to the first day of the month in which the survey results were released. Without the interview dates of the respondents, these gray dots correspond to the highest frequency inflation expectations available from the MSC. Generally, the daily measure of inflation expectations fluctuate around the

Figure 1 – Daily Inflation Expectations from MSC, June 1979 – May 2019

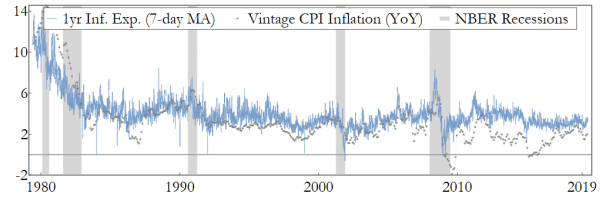
(A) Daily v. Survey



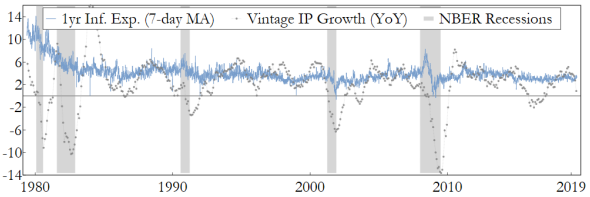
(B) Unemployment Rate



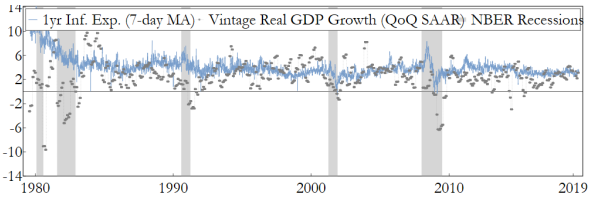
(C) CPI Inflation



(D) IP Growth



(E) Real GDP Growth



Notes: Within each panel, the horizontal axis is daily starting on May 29, 1979 and ending on May 28, 2019. This correspond to the first and last available interview dates of respondents appearing in the June 1979 and May 2019 surveys, respectively. The vertical axis in each panel is percent. In each of these panels, the blue (solid) line is the 7-day moving average of households' one year-ahead expected inflation rate in the Michigan Survey of Consumers. In panel (A), the gray-connected dots is the average expected inflation rate for the survey that released that month, aligned with the first day of each month. These dots are the highest frequency household inflation expectations available without the use of the respondents interview dates. In the remaining panels (B) – (E), the gray-connected dots are the vintage release estimates for each macroeconomic release, aligned to the historical release date the value was published. Vintage refers to the value published in the initial public release, and not the subsequently revised value. These dates are in Table C.1 – Table C.4. The gray bars denote NBER recessions.

survey average, but are noticeable more volatile than the lower frequency series. Additionally, this makes clear the issue of identifying how household inflation expectations respond to macroeconomic data releases, as individuals interviewed in the middle to end of the month necessarily have different information sets than those interviewed in the beginning of the month.[17]

In panels (B) – (E), the vintage release values of the unemployment rate, CPI inflation, IP growth, and real GDP growth are shown along side the daily frequency household inflation expectations.[18] Within each of these panels, the gray-connected dots show the respective release value aligned to the day that the release occurred.[19] Further, it should be noted that in panels

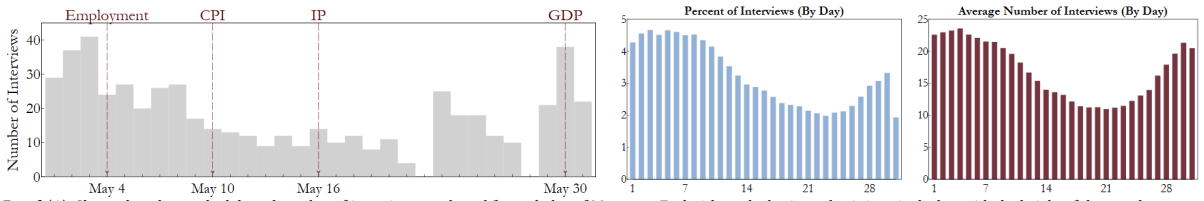
17: Here, this is a standard average of all inflation expectations reported for every respondent interviewed for a particular survey release. This series is *not* the series that the survey publishes itself, which is the series that is also attainable from FRED. The method of constructing that series is discussed here, but more or less the main difference is that survey administrators attempt to account for outliers within the individual responses.

18: Vintage refers to the value published in the initial public release, and not the subsequently revised value.

19: These dates are in Table C.1 – Table C.4.

Figure 2 – Monthly Breakdown of Interviews and Example Release Calendar

(A) Releases in May 2018 **(B)** Distribution of Interviews



Panel (A): Shows the release schedule and number of interviews conducted for each day of May 2018. Each tick on the horizontal axis is a single day, with the height of the gray bar on each day denoting the number of MSC respondents that were interviewed on that day. The release dates shown are for the employment situation (*May 4*), CPI (*May 10*), Industrial Production (*May 16*), and GDP (*May 30*).
Panel (B): In each figure, the horizontal axis are days of the month, starting at 1 and ending at 31. The left figure (*blue*) shows the percent of all interviews ($N = 221{,}845$) occurring on each day for surveys conducted between June 1979 and May 2019. The right figure (*red*) shows the average number of interviews conducted on each day of the month over the same time period.

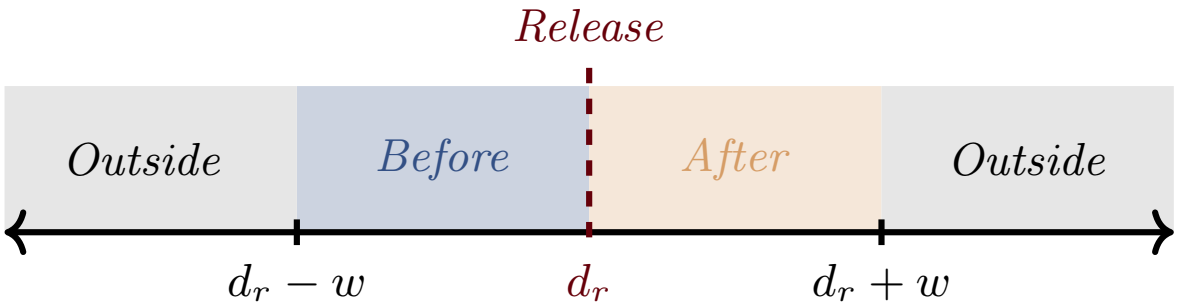
(B) – (D), these vintage values are the estimates of the headline series for the *previous* month, reflecting the standard one month delay in publication of official macroeconomic statistics at this frequency. For real GDP growth in panel (E), the first month of a new quarter reflects the advanced estimates for the preceding quarter, with the two monthly observations following it showing the second and final estimates of that value, respectively. Noteworthy here is the observation that, even at this higher frequency, household inflation expectations follow closely with changes in the unemployment rate and CPI inflation.

Concluding this subsection is Figure 2, showing the distribution of interviews within an average month. Panel (A) shows an example month of interviews and releases occurring in May 2018. Within this panel, the dashed arrows indicate the release dates of the employment situation summary, Consumer Price Index, Industrial Production, and Gross Domestic Product. The gray bars show the number of interviews that were conducted on each day indicated on the horizontal axis. For example, there were no interviews conducted on May 22, 2018 or May 28, 2018. For comparison, panel (B) shows the percent of all interviews within the sample period that were conducted on a particular day of the month (*blue, left*), as well as the average number of interviews conducted on each day (*red, right*). While on average interviews are being conducted every day of any particular month, the extent of these interviews are not always sufficient to estimate *separate* effects for each release, like that of Binder et al. (2022). Instead, pooling the responses of households across multiple releases instances is required.

3.2 Identification Strategy

While the identification strategy employed here is akin to an event study approach, Figure 3 lays out the classification of interviews based on when they occur, relative to the releases of interest. Further, given the aforementioned discussion about the need to pool across multiple

Figure 3 – Timing of Interviews and Economic Releases



Notes: This figure summarizes the event-study style identification strategy employed throughout the remainder of the paper. Starting first with notation, d_r denotes the release date for generic macroeconomic statistic $r \in R$. Here, R would simply be the employment situation, CPI, IP, GDP, and FOMC meetings/calls. Additionally, $\pm w$ generically denotes the number of days around the release date d_r that will be used to classify respondents into one of three potential groups. Individuals interviewed between $[-w, -1]$ days before the release are labeled as the *before* group, while individuals interviewed between $[0, w-1]$ days after the release are labeled as the *after* group. For the latter, the zero refers to interviews conducted on the day of release. Additional details for why this assumption is appropriate is detailed within the text of this section. Interviews that do not satisfy either of these classifications are considered the *outside* group, but it should be noted that interviews may be conducted outside the release window of one release type but within the release window of another. Conversely, releases windows may be overlapping, as highlighted in Table 1.

instances of releases, I will refer to the five releases considered as release *types*, while I will refer to a release occurring at a particular point in time as a release *instance*. In the figure, and in the regression equations to come, I refer to the set of generic macroeconomic releases as R and a particular release type as $r \in R$. Additionally, d_r refers to a release instance of type r.

Now, based off the interview date of respondent j, denoted d_j, I classify each individual as being within one of three different groups, based off the number of days, w, around the release to be considered in the release window. Individuals interviewed more than w days before or $w-1$ days after any release of r, where $d_j - d_r < -w$ and $d_j - d_r > w - 1$ respectively, are considered the *outside* group and denoted by the gray shaded areas in Figure 3.

Next, individuals that are interviewed between 1 and w days preceding the release, where $d_j - d_r \in [-w, 0)$, are denoted as the *before* group. Finally, individuals with interviews occurring between the day of a release and $w-1$ days after, where $d_j - d_r \in [0, w-1]$, are denoted as the *after* group. Throughout this paper, I consider ranges of the release window for one, two, and three days, i.e. $w = 1, 2, 3$, where the one day window would be considered the "tightest" identification window, in the sense that the *before* interviews would all occur the day before a release occurs while the *after* would consist of interviews occurring on the day of a release.

Categorizing the respondents into such groups motivates a rather straightforward regression specification to assess whether household inflation expectations *after* macroeconomic release r are statistically different than household inflation expectations of those interviewed *before* . While it will be modified in various ways throughout the remainder of the paper to address specific questions of interest, the baseline regression equation takes the following form:[20]

20: While not of particular interest within the main text, Table A.1 shows estimates of (1). This specification is

$$\pi_j^e = \boldsymbol{x}_j'\boldsymbol{\gamma} + \sum_{r \in R} \left(\beta_r \mathbb{I}_r^b + \alpha_r \mathbb{I}_r^a \right) + \varepsilon_j \tag{1}$$

Where the dependent variable, π_j^e is individual j's expected one-year ahead inflation rate, $\boldsymbol{x}_j$ denotes the generic vector of control variables. The summation is over each macroeconomic release of interest, with each release type getting a set of indicators to denote whether individual j was interviewed *before* ($\mathbb{I}_r^b = 1$) any release of r, or interviewed *after* ($\mathbb{I}_r^a = 1$). For those not interviewed outside the release window, both of these indicators are zero. To keep notation within (1) light, I have suppressed references to interview dates d_j, release dates d_r, and the window around the release date w, but the specification follows directly from Figure 3.

Thus, to discern whether there is a statistically significant adjustment in household inflation expectations in response to a release of r, our interest is in testing $H_0 : \alpha_r - \beta_r = 0$, with $\hat{\alpha}_r - \hat{\beta}_r$ measuring the difference in inflation expectations after the release, using the inflation expectations of interviews conducted before the release as the control group. Further, it should be noted that for both groups, the reference group are interviews conducted outside of the release window. Finally, it benefits to reiterate that these coefficients are being pooled across time, so they correspond to the conditional shift in expectations using all the historical releases of r, not estimating separate responses for each *instance* of r being released. To clarify with an example, one release that will be of interest is CPI, where the headline estimate of inflation is published. The coefficients on the indicators in (1) are estimating the conditional mean shift due to *any* CPI release, not the CPI release for a particular point in time. While the latter is certainly of interest, this pooling is done to account for the previous discussion of the density of interviews conducted around historical releases.

While there are certainly numerous macroeconomic releases that could be of interest, I focus on five in particular. These are the Employment Situation Summary (*ESS*), Consumer Price Index (*CPI*), Industrial Production (*IP*), Gross Domestic Product (*GDP*), and Federal Open Market Committie (*FOMC*) meetings/calls. The historical dates for each of these are included Table C.1 - Table C.5. Between June 1979 and May 2019, theses include 480 monthly *ESS*, *CPI*, and *IP* releases. For *GDP*, there are 239 advanced/second and 240 final estimate releases, where the latter is different because the sample starts in the last month of 1979:Q2. Finally, there are 459 FOMC meetings and calls, with the February 1982 – December 2014 dates coming from

not of particular interest because it does not incorporate characteristics of the release, thus not measuring any of the information that is being conveyed.

Nakamura and Steinsson (2018) and Gürkaynak et al. (2005), with the additional events needed to span June 1979 – May 2019 done by this paper.

Returning to the method of classifying interviews as having been conducted *before* ,*after* , or *outside* releases, some institutional details for these releases serve as additional justification for the hard cutoff at the day of the release. Ideally, if one were to not only know the day of the respondents interview but also the start and end times, then one could unambiguously distinguish between before and after interviews on the days of releases.[21] However, since the data does not include any method for making a distinction between interviews conducted before or after on the day of release, I rely on the historical times of these releases to motivate this assumption.

Starting with the two reports published by the Bureau of Labor Statistics (BLS), which are the *ESS* and *CPI*, both reports are publicly released at 8:30 am ET. Following from the fact that the MSC includes respondents from various time zones across the U.S., then including all interviews on the day of these releases in the *after* is arguably trivial.[22] The same justification applies for Bureau of Economic Analysis (BEA) releases of *GDP*, which also are made publicly available at 8:30am ET. Finally, the Borad of Governors of the Federal Reserve have historical released *IP* sometime between 9:15 – 9:30 am ET, at least since the start of the sample in 1979.[23]

This assertion is much less clear for FOMC meetings and calls because historically these have occurred later in the day. Thus, while it is certainly possible that some of these interviews were conducted before the meetings or calls actually took place, I maintain the same assertion for consistency with the other releases. However, these events present the unique aspect in that historically they have frequently occurred over multiple days. In such instances, I consider the *before* group to be interviews conducted before the *first* event day, and anyone interviewed on one of the days the event occurred to be in the *after* group. In other words, the "release date" for FOMC meetings/calls would be considered *every* day where events were held, and then the *after* group is calculated from the *last* day of the meetings and from the *first* meeting day for the *before* group.

Another potential issue with the identification strategy is the fact that historically many of these releases have occurred within very close proximity to one another. Thus, incorporating

21: Interestingly, an additional piece of information that is included in the ICPSR data but not the MSC data is the duration of each interview, but it does not include explicitly what time the interview began.
22: The only time where one of these reports were released later in the day occurred for employment situation on November 5, 1998, where the report was released at 1:30 pm ET because estimates were leaked on the internet a day early, initially scheduled to release the following morning at 8:30 am ET.
23: These times can be found on FRASER here and here.

Table 1 – Overlap Between Historical Releases, June 1979 – May 2019

Release	Overlaps With	Same Day (%)	One Day (%)	Two Days (%)	Three Days (%)	Average Days (#)
Consumer Price Index	Employment Situation	–	0.2	–	–	11.1
	Industrial Production	21.3	22.5	9.4	5.4	3.4
	Gross Domestic Product	2.7	4.6	4.8	2.5	9.2
	FOMC Meetings/Calls	10.0	9.8	3.5	5.0	7.2
Employment Situation	Industrial Production	–	–	–	–	10.6
	Gross Domestic Product	0.2	2.3	5.6	1.9	9.1
	FOMC Meetings/Calls	1.9	1.0	5.0	16.0	8.1
Industrial Production	Gross Domestic Product	0.4	1.0	2.5	2.1	10.1
	FOMC Meetings/Calls	5.4	10.0	3.8	5.6	7.4
Gross Domestic Product	FOMC Meetings/Calls	5.6	7.1	6.3	9.4	6.9

Notes: For each release considered (**Release**), the columns *Same Day, One Day, Two Days,* and *Three Days* shows the percent of another release (**Overlaps with**) that occurred that number of days apart. Only occurrences within three days of each other are shown, but the average in the last column is calculated using all of differences in release dates. This percent is calculated with respect to the minimum of the absolute value of the difference between releases dates, so that order does not matter. The release dates span from June 1979 to May 2019. The specific days for each release date are included in appendix tables Table C.1 – Table C.5. The full sample includes 480 releases of CPI, employment situation, industrial production and GDP, as well as 459 FOMC meetings or conference calls.

the release window indicators in (1) separately may be problematic. To alleviate this concern, Table 1 shows the historical frequency at which each of these releases occurred within zero to three days of each other. In the table, the first column contains a grouping for each of the four data releases, and the second column fixes the remaining releases. The middle set of columns shows the percent of historical releases that were separated by the number of days at the top of the column. For example, the first row shows that, between June 1979 and May 2019, the Consumer Price Index and Employment Situation Summary never released on the same day, two days apart, or three days apart. They were released one day about 0.2% of the time, which corresponds to only a single instance of such difference for the 480 monthly releases of each.

The last column in the table shows the average number of days between each of the releases. This average includes releases occurring more than seven days apart, but are excluded from the table. For the most part, this frequency of overlapping releases motivates the consideration of the one, two, and three day windows around releases stated previously. The combination that most likely would raise concern is the average number of days between CPI and IP releases being 3.3 days. Results presented are robust to various methods of accounting for this, including interactions of the release indicators, incorporating general release window overlap fixed effects, etc. Thus, the results utilizing the separate indicators for both release types are shown, as their interpretation and discussion are much more convenient.

3.3 Details on Control Variables

As summarized in Table 2, I consider a wide variety of control variables throughout the paper. For consistency, these will be the controls used for all of the main text results, albeit with

Table 2 – Definitions for each Group of Control Variables

Base Controls	
Demographics	Income (*tertiles by survey*), age (*Under 30, 30-64, 65 or Older*), region (*Midwest, Northeast, South, West*), gender, martial status, number of household members, college degree, interviewed before
Year and Month	Indicators for survey month and year
President and Fed Chair	Interacted indicators for combintations of U.S. Presidents (*Carter – Trump*) and Federal Reserve Chairs (*Miller – Yellen*)
+ Sentiment and News	
Sentiment Responses	Responses for each of five questions included in MSCs index of consumer sentiment
News Heard	Reported hearing no news, favorable, unfavorable, or mixed
+ Macro Aggregates	
Aggregates	Lagged close to close level change in S&P 500, and lagged Aruoba-Diebold-Scotti (ADS) Business Conditions Index

Notes: All variables are considered from June 1979 to May 2019. The demographic variables, sentiment responses, and news heard come directly from Michigan Survey of Consumers (MSC), with some minor edits where necessary. Income is broken up into a group indicator for which tertile their income falls into, relative to the incomes reported in the survey for which their responses are included. This approach is also useful because the format of the income question has changed over time, varying from a bracketed response to individuals recording their income themselves. Similarly, age is broken up in a categorical variable for *under 30, 30 to 64, 65 or older.* For region, the survey refers to the *midwest* as *north central*, but each of their regions follow the census grouping of states, so I refer to it as the former. Number of household members refers to the sum of the surveys number of adults and number of kids within the household questions. Interviewed before is an indicator for whether the respondent was randomly selected to be re-interviewed 6 months after their initial interview.
Section Appendix B: Contains additional details, including the text of questions appearing in the survey and possible responses. This appendix also contains figures showing the demographic composition of survey respondents and responses included in the *sentiment and news* grouping.
Section Appendix C: Contains figures documenting the demographic composition and sentiment/news responses within the release windows.

some of the later sub-samples having additional control variables that could be included. The demographic set of control variables all come from the MSC directly, with some minor changes discussed in the footnote of Table 2. Additionally, given extensive discussion of interview dates thus far, one might naturally expect that the year and month fixed effects should alternatively be specified for interview year and month. While this produces entirely analogous results given that the vast majority of interviews do in fact occur within the same month as the survey they are being interviewed for, the proper "clustering" is on the survey in which their responses would appear, as by the end of the month the previous results have already been released. Finally, to control for the different policy regimes observed over the sample, I include interacted categorical variables for the U.S. president and Chair of the Federal Reserve on the interview date. The sample begins with President Jimmy Carter and Chair William Miller, and ends with President Donald Trump and Chair Janet Yellen.[24]

As the literature has established, household inflation expectations exhibit noticeable idiosyncratic variation driven by a range of factors, for example how exposed the individual is to prices that are more volatile. While the survey does not explicitly document purchasing

24: Even though the coefficients for these are statistically significant as controls variables, I do not find that households have historically responded differently to these releases under the different combinations. This was done by interacting the release window indicators with the interaction of the policy regime indicators, and yielded estimates that were not statistically different from zero.

behavior from the respondents, incorporating some qualitative questions that the survey asks can help account for at least some of this idiosyncratic portion. Theses are their responses to the surveys "sentiment" questions, as well as the news the individual has reported hearing. One may immediately think that incorporation of these responses are problematic, however probit estimates suggest that there is not a statistically significant difference in the probability of a particular response after the release, with respect to individuals interviewed before.[25] For the sentiment questions, these correspond to the five questions included in the MSC headline Index of Consumer Sentiment (ICS). Two questions are backward looking, in the sense that they ask the individual whether something is better, worse, or the same as it was a year ago, whereas the remaining three questions are forward looking, asking the respondent how they expect things to change over the next year. For each question, the responses are included as indicators.

The news heard variable is constructed from two news related questions that the survey asks respondents. The respondent is prompted as to whether or not, in the last few months, they have heard of any favorable or unfavorable changes in business conditions, and what exactly did they hear. Depending on the general thematic content of the respondents answer, the *interviewer* classifies the response into either a favorable or unfavorable response, and then into one of four categories, where each category has approximately 10 specific reasons the answer could fall under. A full breakdown of the categories and possible reasons is provided in Table B.3.

As the interviewer records the respondents answer to this question two times, the survey records these into two news variables, which I use to construct the news heard variable. Regardless of the specific reasons recorded, I classify a respondent as having heard *favorable* news if either/both of their responses in the *news1/news2* variables are classified as being favorable, and neither of the responses would be classified as *unfavorable*. For example, a respondent may report hearing favorable news when initially prompted, which would be recorded in the *news1* variable, and then report hearing no additional news, so that *news2* would be coded as 0, denoting that the respondent did not mention any news during the second prompt. This would be classified as *favorable* news heard in my variable. An analogous method for classifying *unfavorable* news is used. In the case where respondents give conflicting responses in the *news1/news2*, for example by reporting having heard favorable price news and

25: For details on the full text and possible responses to the sentiment question, as well as the surveys news related variables, see Section Appendix B. For a graphical breakdown of how these responses change within the ±3 window around releases, see the figures at the end of Section Appendix C. Appendix with probit estimates is forthcoming.

unfavorable government/defense news, then their news heard variable would be classified as *mixed*. Finally, respondents are classified as having heard no news if both *news1/news2* reflect such a response.

Finally, I additionally include certain macroeconomic aggregates. Based on the date of the respondents interview, this set of control variables includes the one day lagged values of the close to close change in the S&P 500, and the lagged value of the ADS business conditions index. These should help control for any additional widespread information that could account for changes in inflation expectations, but at the same time not reflect the realized value of macroeconomic statistics on their release dates.

4 Inflation Expectations Around Data Releases

This section can be considered the baseline results of this paper, wherein the first portion discusses and presents results for the full June 1979 – May 2019 sample, while the second portion presents results for a sub-sample where additional FOMC meeting information is availble.

4.1 Response to Releases: June 1979 – May 2019

As indicated in Section 1, I consider how household inflation expectations respond to macroeconomic data releases when incorporating some characteristics of the releases, as to account for the information that is being conveyed. I refer to these as the *sign* and *size* of the release, wherein each is referring to the relationship between the new estimates of the releases headline series with respect to its previous estimates. Further, by modifying (1) in Section 3, these characteristics are included as:

$$\pi_j^e = \boldsymbol{x}_j'\boldsymbol{\gamma} + \sum_{r\in R}\left(\beta_r^+\mathbb{I}_r^{b+} + \alpha_r^+\mathbb{I}_r^{a+} + \beta_r^-\mathbb{I}_r^{b-} + \alpha_r^-\mathbb{I}_r^{a-}\right) + \varepsilon_j \tag{2}$$

$$\pi_j^e = \boldsymbol{x}_j'\boldsymbol{\gamma} + \sum_{r\in R}\Delta_r^v\left(\beta_r\mathbb{I}_r^{b} + \alpha_r\mathbb{I}_r^{a}\right) + \varepsilon_j \tag{3}$$

$$\pi_j^e = \boldsymbol{x}_j'\boldsymbol{\gamma} + \sum_{r\in R}\Delta_r^v\left(\beta_r^+\mathbb{I}_r^{b+} + \alpha_r^+\mathbb{I}_r^{a+} + \beta_r^-\mathbb{I}_r^{b-} + \alpha_r^-\mathbb{I}_r^{a-}\right) + \varepsilon_j \tag{4}$$

Where the dependent variable, π_j^e is individual j's expected one-year ahead inflation rate and $\boldsymbol{x}_j$ denotes a vector of control variables from Table 2. The summation is over a set of indicators for each macroeconomic release of interest, where interviews classified as *before* any release of

r have $\mathbb{I}_r^b = 1$, while interviews classified as *after* have $\mathbb{I}_r^a = 1$. For the remaining interviews, these indicators are both zero. For this subsection, only indicators for being interviewed around FOMC meetings/calls are used, with further discussion of this occurring in the next subsection. Starting with (2), this further decomposes the *before* and *after* indicators into whether their interviews were conducted around a release where the headline series was *higher* than the previous release ($\mathbb{I}_r^{b+} = 1$ or $\mathbb{I}_r^{a+} = 1$), or whether it was *lower* or the same as the previous release ($\mathbb{I}_r^{b-} = 1$ or $\mathbb{I}_r^{a-} = 1$). Together, interest is in whether household inflation expectations respond to positive or negative releases, which are $\hat{\alpha}_r^+ - \hat{\beta}_r^+$ and $\hat{\alpha}_r^- - \hat{\beta}_r^-$, respectively.

For (3), $\Delta_r^v = v_r^{new} - v_r^{last}$ is the *size* of the change in the headline series for release r. This would correspond to the change in the growth rate for GDP and IP, inflation rate for CPI, and unemployment rate. Further, in all cases I use the real-time vintages of the series, not the historically revised values. This ensures that the change accurately reflects what would have been observed around each release window. Here, the *before* and *after* indicators follow their original definition, thus estimating how household inflation expectations respond to a 1 percentage point (p.p.) or annualized percentage point (a.p.p.) change in each headline series. This corresponds to $\Delta_r^v(\hat{\alpha}_r - \hat{\beta}_r)$. Finally, (4) combines both of the preceding equations, which will estimate whether household inflation expectations respond more increases or decreases in each headline series, accounting for the *size* of each type, respectively.

Table 3 shows estimates of (2). The columns of the table with **bold** names correspond to the set of control variables included, from Table 2. Under each of these, the columns with $w = 1, 2, 3$ indicate the number of days around the releases used to categorize interviews as having occurred before or after a release. The aforementioned releases are denoted by the rows, under which the rows labeled *Lower/Same as Previous* denotes instances where the headline series is lower or the same as the previously known value, whereas *Higher than Previous* denotes the converse. These correspond to the negative and positive superscripts, respectively.

The key takeaway from considering the sign of the release is that, regardless of how much lower the headline CPI inflation rate is with respect to its previous value, inflation expectations are lower after the release. To make clear the magnitude for these results, and the forthcoming results, consider that average household inflation expectations were 1 percent for those interviewed before the CPI release. The estimates shown in the *lower/same as previous* row would indicate that household inflation expectations decrease by $0.1 - 0.2$ percentage points, on average being $0.8 - 0.9$ percent over the next year. Further, under the "tightest" identification window ($w = 1$), extending the set of control variables changes estimates only slightly, suggesting that a lower CPI inflation rate decreases household inflation expectations

Table 3 – Do Household Inflation Expectations Respond to the Direction of Change in the Headline Series of each Release?

Dep Var: 1yr Inf. Exp.	**Base Controls**			**+ Sentiment and News**			**+ Macro Aggregates**		
Coef:$(\hat{\alpha}_r^- - \hat{\beta}_r^-)$ and $(\hat{\alpha}_r^+ - \hat{\beta}_r^+)$	$w=1$	$w=2$	$w=3$	$w=1$	$w=2$	$w=3$	$w=1$	$w=2$	$w=3$
Unemployment Rate									
Lower/Same as Previous ($\mathbb{I}_r^- = 1$)	-0.036	-0.028	-0.005	-0.007	-0.008	0.035	-0.005	-0.009	0.031
	(0.056)	(0.043)	(0.036)	(0.061)	(0.047)	(0.039)	(0.061)	(0.047)	(0.039)
Higher than Previous ($\mathbb{I}_r^+ = 1$)	-0.110	-0.119*	-0.086	-0.089	-0.100	-0.065	-0.093	-0.106	-0.072
	(0.095)	(0.071)	(0.059)	(0.103)	(0.077)	(0.064)	(0.102)	(0.077)	(0.064)
CPI Inflation (MoM)									
Lower/Same as Previous ($\mathbb{I}_r^- = 1$)	-0.195**	-0.062	-0.111**	-0.182*	-0.025	-0.098*	-0.185*	-0.028	-0.100*
	(0.088)	(0.066)	(0.055)	(0.094)	(0.070)	(0.059)	(0.094)	(0.070)	(0.059)
Higher than Previous ($\mathbb{I}_r^+ = 1$)	0.088	0.080	0.040	0.039	0.004	-0.022	0.044	0.008	-0.019
	(0.092)	(0.071)	(0.060)	(0.099)	(0.077)	(0.065)	(0.099)	(0.077)	(0.065)
IP Growth (MoM)									
Lower/Same as Previous ($\mathbb{I}_r^- = 1$)	0.050	0.011	-0.003	0.114	0.069	0.028	0.113	0.066	0.025
	(0.087)	(0.065)	(0.055)	(0.093)	(0.070)	(0.059)	(0.093)	(0.070)	(0.059)
Higher than Previous ($\mathbb{I}_r^+ = 1$)	0.036	-0.010	0.005	0.054	0.035	0.040	0.059	0.040	0.044
	(0.089)	(0.069)	(0.057)	(0.098)	(0.075)	(0.062)	(0.098)	(0.075)	(0.062)
Real GDP Growth (QoQ)									
Lower/Same as Previous ($\mathbb{I}_r^- = 1$)	0.015	0.101	0.113*	0.012	0.088	0.076	0.018	0.097	0.085
	(0.097)	(0.073)	(0.061)	(0.104)	(0.078)	(0.065)	(0.104)	(0.078)	(0.065)
Higher than Previous ($\mathbb{I}_r^+ = 1$)	0.144	0.018	-0.002	0.179*	0.005	0.006	0.177*	0.005	0.007
	(0.090)	(0.066)	(0.055)	(0.098)	(0.072)	(0.059)	(0.098)	(0.072)	(0.059)
FOMC Meetings/Calls (No Sign)									
Diffference	-0.004	-0.018	0.011	0.009	-0.010	0.021	0.008	-0.013	0.020
	(0.061)	(0.048)	(0.041)	(0.066)	(0.052)	(0.045)	(0.066)	(0.052)	(0.045)
Observations	221,845	221,845	221,845	184,930	184,930	184,930	184,930	184,930	184,930
Adjusted R^2	0.086	0.086	0.086	0.119	0.119	0.119	0.119	0.119	0.119

Estimates of $(\hat{\alpha}_r^- - \hat{\beta}_r^-)$ and $(\hat{\alpha}_r^+ - \hat{\beta}_r^+)$ from (2): $\pi_j^e = \boldsymbol{x}_j'\boldsymbol{\gamma} + \sum_{r\in R}(\beta_r^+\mathbb{I}_r^{b+} + \alpha_r^+\mathbb{I}_r^{a+} + \beta_r^-\mathbb{I}_r^{b-} + \alpha_r^-\mathbb{I}_r^{a-}) + \varepsilon_j$

Notes: $^{***}p < 0.01, ^{**}p < 0.05, ^{*}p < 0.10$. Robust standard errors are in parentheses. Sample is for June 1979 – May 2019. Bolded column headings are the control variables included, defined in Table 2, and the columns labeled with $w = 1, 2, 3$ denote the number of days before and after used to define the release window. Dependent variable is individuals reported on year ahead expected inflation rate from Michigan Survey of Consumers (MSC), excluding individuals with reported expected inflation rate greater than 20 percent in absolute value. For releases of the employment situation, consumer price index, industrial production, gross domestic product, and FOMC meetings/calls, the headline series used is in **bold**. The rows labeled *Lower/Same as Previous* denotes instances where the headline series is lower or the same as the previously known value, whereas *Higher than Previous* denotes the converse. These correspond to the negative and positive superscripts, respectively.

by about 0.18 percentage points. However, an interesting dichotomy presents itself when considering not the sign of the release, but instead the size of the release, corresponding to (3) and shown in Table 4.

Changes in the headline unemployment rate, as indicated by the estimates in the first row of Table 4, prompt significant adjustment in household inflation expectations. Each row shows the estimated percentage point response of household inflation expectations for each percentage point (p.p.) or annualized percentage point (a.p.p.) change in the corresponding headline series. Here, an increase in the unemployment rate from 7 to 8 percent suggests that household inflation expectations decrease by around 0.6 percentage points. Additionally, this estimated

Table 4 – Does the Size of the Change Matter?

Dep Var: 1yr Inf. Exp.	**Base Controls**			**+ Sentiment and News**			**+ Macro Aggregates**		
Coef: $\Delta_r^v(\hat{\alpha}_r - \hat{\beta}_r)$	$w=1$	$w=2$	$w=3$	$w=1$	$w=2$	$w=3$	$w=1$	$w=2$	$w=3$
Unemployment Rate ($\Delta_r^v = 1$ p.p.)	-0.613**	-0.553**	-0.251	-0.641**	-0.569**	-0.320	-0.660**	-0.585**	-0.338*
	(0.294)	(0.220)	(0.182)	(0.316)	(0.236)	(0.196)	(0.316)	(0.236)	(0.196)
CPI Inflation (MoM, $\Delta_r^v = 1$ a.p.p.)	0.031	0.020	0.012	0.023	-0.000	-0.002	0.024	0.001	-0.002
	(0.020)	(0.015)	(0.013)	(0.022)	(0.016)	(0.014)	(0.022)	(0.016)	(0.014)
IP Growth (MoM, $\Delta_r^v = 1$ a.p.p.)	-0.011	-0.008	-0.003	-0.012	-0.009	-0.003	-0.011	-0.008	-0.002
	(0.008)	(0.006)	(0.005)	(0.008)	(0.006)	(0.005)	(0.008)	(0.006)	(0.005)
Real GDP Growth (QoQ , $\Delta_r^v = 1$ a.p.p.)	0.033	0.041	0.024	0.051	0.053	0.037	0.050	0.053	0.038
	(0.048)	(0.035)	(0.028)	(0.052)	(0.038)	(0.030)	(0.052)	(0.038)	(0.030)
FOMC Meetings/Calls (No Change)	0.002	-0.004	0.020	0.015	0.005	0.025	0.014	0.003	0.025
	(0.061)	(0.048)	(0.041)	(0.065)	(0.051)	(0.044)	(0.065)	(0.051)	(0.044)
Observations	221,845	221,845	221,845	184,930	184,930	184,930	184,930	184,930	184,930
Adjusted R^2	0.086	0.086	0.086	0.119	0.119	0.119	0.119	0.119	0.119

Estimates of $\Delta_r^v(\hat{\alpha}_r - \hat{\beta}_r)$ from (3): $\pi_j^e = \boldsymbol{x}_j'\boldsymbol{\gamma} + \sum_{r\in R} \Delta_r^v(\beta_r \mathbb{I}_r^b + \alpha_r \mathbb{I}_r^a) + \varepsilon_j$

Notes: $^{***}p < 0.01, ^{**}p < 0.05, ^{*}p < 0.10$. Robust standard errors are in parentheses. Sample is for June 1979 – May 2019. Bolded column headings are the control variables included, defined in Table 2, and the columns labeled with $w = 1, 2, 3$ denote the number of days before and after used to define the release window. Dependent variable is individuals reported on year ahead expected inflation rate from Michigan Survey of Consumers (MSC), excluding individuals with reported expected inflation rate greater than 20 percent in absolute value. For releases of the employment situation, consumer price index, industrial production, gross domestic product, and FOMC meetings/calls, the headline series used is in **bold**. Each row shows the estimated percentage point response of household inflation expectations for each percentage point (p.p.) or annualized percentage point (a.p.p.) change in the corresponding headline series.

decrease is nearly 3 times larger than the estimated decrease after CPI releases in the previous table.

Together, Table 3 and Table 4 present two important conclusions. First, for the releases considered, significant responses are only to the information contained in CPI inflation and unemployment rate releases. As this coincides with the Federal Reserves dual mandate, consistent and subsequent policy actions indirectly affect household inflation expectations. Second, the relevant piece of information from different releases may themselves differ. For example, households adjusting expectations depending on the sign but not the size of CPI releases is consistent with the relevant piece of information being the *direction* of changes, not how much it changes by.

This conclusion is further reinforced when considering the adjustment in household inflation expectations to the size of increases and decrease of the releases separately, corresponding to (4) and shown in Table 5. The estimates under CPI inflation indicate that, even when considering possible asymmetric adjustment to positive/negative changes in the headline rate, the size of these changes is not what is relevant. The relevant portion, on average, is just whether the information in the release conveys that the headline inflation rate was lower than previously estimated. In contrast, the estimates in the first two rows of the table indicate that is is *increases* in the unemployment rate that prompt significant decreases in household inflation expectations.

Table 5 – Do Households Respond More to Increases or Decreases in each Headline Series?

Dep Var: 1yr Inf. Exp.	**Base Controls**			**+ Sentiment and News**			**+ Macro Aggregates**		
Coef:$\Delta_r^v(\hat{\alpha}_r^+ - \hat{\beta}_r^+)$ and $\Delta_r^v(\hat{\alpha}_r^- - \hat{\beta}_r^-)$	$w=1$	$w=2$	$w=3$	$w=1$	$w=2$	$w=3$	$w=1$	$w=2$	$w=3$
Unemployment Rate (Δ_r^v = 1 p.p.)									
Decrease from Previous	-0.421	-0.443	-0.273	-0.494	-0.467	-0.469*	-0.515	-0.473	-0.469*
	(0.387)	(0.291)	(0.240)	(0.415)	(0.310)	(0.256)	(0.416)	(0.310)	(0.256)
Increase from Previous	-0.785*	-0.643**	-0.243	-0.770*	-0.650*	-0.213	-0.787*	-0.674*	-0.241
	(0.432)	(0.322)	(0.265)	(0.463)	(0.345)	(0.285)	(0.462)	(0.345)	(0.285)
CPI Inflation (MoM, Δ_r^v = 1 a.p.p.)									
Decrease from Previous	0.013	0.007	0.011	0.007	-0.009	0.004	0.006	-0.009	0.004
	(0.028)	(0.021)	(0.018)	(0.030)	(0.023)	(0.019)	(0.030)	(0.023)	(0.019)
Increase from Previous	0.054*	0.035	0.014	0.043	0.010	-0.008	0.045	0.012	-0.007
	(0.029)	(0.022)	(0.018)	(0.031)	(0.024)	(0.020)	(0.031)	(0.024)	(0.020)
IP Growth (MoM, Δ_r^v = 1 a.p.p.)									
Decrease from Previous	-0.016	-0.007	-0.003	-0.020*	-0.013	-0.007	-0.020*	-0.013	-0.007
	(0.011)	(0.008)	(0.007)	(0.012)	(0.009)	(0.007)	(0.012)	(0.009)	(0.007)
Increase from Previous	-0.008	-0.007	-0.002	-0.006	-0.005	0.002	-0.005	-0.004	0.003
	(0.011)	(0.009)	(0.007)	(0.012)	(0.009)	(0.008)	(0.012)	(0.009)	(0.008)
Real GDP Growth (QoQ, Δ_r^v = 1 a.p.p.)									
Decrease from Previous	-0.047	0.012	0.010	-0.010	0.050	0.049	-0.017	0.044	0.046
	(0.066)	(0.048)	(0.039)	(0.073)	(0.053)	(0.042)	(0.073)	(0.053)	(0.042)
Increase from Previous	0.134*	0.079	0.043	0.126*	0.056	0.023	0.134*	0.063	0.028
	(0.069)	(0.051)	(0.041)	(0.073)	(0.055)	(0.043)	(0.073)	(0.055)	(0.043)
FOMC Meetings/Calls (No Sign/Size)									
Diffference	0.004	-0.005	0.018	0.016	0.003	0.025	0.015	-0.001	0.023
	(0.061)	(0.048)	(0.041)	(0.066)	(0.052)	(0.045)	(0.066)	(0.052)	(0.045)
Observations	221,845	221,845	221,845	184,930	184,930	184,930	184,930	184,930	184,930
Adjusted R^2	0.086	0.086	0.086	0.119	0.119	0.119	0.119	0.119	0.119

Estimates of $\Delta_r^v(\hat{\alpha}_r^- - \hat{\beta}_r^-)$ and $\Delta_r^v(\hat{\alpha}_r^+ - \hat{\beta}_r^+)$ from (4): $\pi_j^e = \boldsymbol{x}_j'\boldsymbol{\gamma} + \sum_{r\in R}\Delta_r^v(\beta_r^+\mathbb{I}_r^{b+} + \alpha_r^+\mathbb{I}_r^{a+} + \beta_r^-\mathbb{I}_r^{b-} + \alpha_r^-\mathbb{I}_r^{a-}) + \varepsilon_j$

Notes: $^{***}p<0.01, ^{**}p<0.05, ^{*}p<0.10$. Robust standard errors are in parentheses. Sample is for June 1979 – May 2019. Bolded column headings are the control variables included, defined in Table 2, and the columns labeled with $w = 1, 2, 3$ denote the number of days before and after used to define the release window. Dependent variable is individuals reported on year ahead expected inflation rate from Michigan Survey of Consumers (MSC), excluding individuals with reported expected inflation rate greater than 20 percent in absolute value. For releases of the employment situation, consumer price index, industrial production, gross domestic product, and FOMC meetings/calls, the headline series used is in **bold**. The rows labeled *Decrease from Previous* show the estimated percentage point response of household inflation expectations when the headline series decreases by one percentage point (p.p.) or annualized percentage point (a.p.p.) with respect to its previously known value. In these estimates, the negative sign is included in Δ_r^v, so the estimates shown in these rows *should* be multiplied by -1. Anagously, rows labeled with *Increase from Previous* show the estimated response when the headline series increases by one p.p. or a.p.p. relative to its previously known value.

4.2 Policy Actions of FOMC: September 1982 – May 2019

As the results presented in Table 3 – Table 5 indicate, a significant portion of the full sample corresponds to historical FOMC meetings and calls without *explicit* intended policy changes. Indeed, a tremendous amount of academic effort has been spent dating pre-1990s monetary policy actions of the FOMC. Such efforts are necessary because the FOMC did not begin officially announcing their intended policy rate changes until February 1994.[26] As noted in Gürkaynak et al. (2005), the announcement of official policy actions were done via press release after every meeting and every policy action. While these were released at differing times throughout the

26: Reuters provides an excellent historical accounting of steps the Federal Reserve has taken to increase the transparency of their deliberations.

first year they were issued, since 1995 they have mostly occurred around 2:15pm ET. Although this practice began in 1994, the Board of Governors of the Federal Reserve does provide these intended changes starting on July 13, 1990.[27] Attributing intended policy actions to FOMC meetings before this date has been left to researchers, with a wide-variety of methods being employed.

As such, the identification strategy employed throughout this paper requires dating these pre-1990s policy actions to specific FOMC meetings or calls. This is done by accumulating historical FOMC information from primarily three sources. First, as briefly discussed in Section 3, I start with the February 1982 – December 2014 FOMC event dates from Nakamura and Steinsson (2018), whom themselves extended the original dates of January 1990 – December 2004 from Gürkaynak et al. (2005). Ensuring the continuity of such a tradition, this paper further expands the dates to span June 1979 – May 2019. For meetings occurring during or after 1990, the intended policy actions are taken directly from the aforementioned source on the Boards' website.

From Thornton (2005), I take the intended policy changes spanning September 27, 1982 to July 13, 1990. For those unfamiliar, it is of interest to mention two points that subsequently have a direct impact on the results presented in this section. First, the paper relies heavily on a narrative-based strategy, wherein a large portion of the changes must be identified via meeting/call transcripts. In particular, this applies most prominently to inter-meeting adjustments, where the committee decides to change the federal funds rate throughout the course of typical deliberation. With, at moments, conflicting opinions and rate suggestions, their exists some subjectivity in this strategy.[28]

With these details in mind, the results of this section correspond to re-estimating equations (2) – (4) from the full sample discussion, but truncating the interview and release dates to correspond to September 1982 – May 2019. Therefore, the corresponding tables for the relevant comparison in the previous section occur in the same order as they appear here. Table 6 shows estimates of (2) for September 1982 – May 2019. Notably, even over the later sub-sample considered the results suggest household inflation expectations decrease when CPI inflation releases were lower than previously estimated, albeit less robust than the first set of results. For FOMC meetings, estimates in the last rows of the table suggest that inflation expectations

27: Those intended policy changes are available under the Boards' Open Market Operations portion of their website.

28: However, the fact that the paper additional includes specific reasoning for *each* of the intended policy changes dated is worth mentioning and applauding.

Table 6 – Sign of Release Changes, September 1982 – May 2019

Dep Var: 1yr Inf. Exp.	**Base Controls**			**+ Sentiment and News**			**+ Macro Aggregates**		
Coef:$(\hat{\alpha}_r^- - \hat{\beta}_r^-)$ and $(\hat{\alpha}_r^+ - \hat{\beta}_r^+)$	$w=1$	$w=2$	$w=3$	$w=1$	$w=2$	$w=3$	$w=1$	$w=2$	$w=3$
Unemployment Rate									
Lower/Same as Previous ($\mathbb{I}_r^- = 1$)	-0.027	-0.036	-0.007	-0.015	-0.033	0.028	-0.016	-0.035	0.024
	(0.056)	(0.043)	(0.036)	(0.060)	(0.046)	(0.038)	(0.060)	(0.046)	(0.038)
Higher than Previous ($\mathbb{I}_r^+ = 1$)	-0.147	-0.168**	-0.140**	-0.126	-0.151*	-0.125*	-0.129	-0.157**	-0.130**
	(0.097)	(0.073)	(0.060)	(0.105)	(0.079)	(0.065)	(0.105)	(0.079)	(0.065)
CPI Inflation (MoM)									
Lower/Same as Previous ($\mathbb{I}_r^- = 1$)	-0.176**	-0.073	-0.136**	-0.146	-0.028	-0.113*	-0.148	-0.031	-0.115*
	(0.087)	(0.066)	(0.055)	(0.094)	(0.070)	(0.059)	(0.094)	(0.070)	(0.059)
Higher than Previous ($\mathbb{I}_r^+ = 1$)	0.162*	0.123*	0.089	0.131	0.061	0.039	0.133	0.062	0.039
	(0.092)	(0.071)	(0.060)	(0.099)	(0.076)	(0.065)	(0.099)	(0.076)	(0.065)
IP Growth (MoM)									
Lower/Same as Previous ($\mathbb{I}_r^- = 1$)	0.021	0.019	-0.015	0.087	0.082	0.020	0.089	0.081	0.019
	(0.086)	(0.064)	(0.055)	(0.093)	(0.069)	(0.059)	(0.093)	(0.069)	(0.059)
Higher than Previous ($\mathbb{I}_r^+ = 1$)	0.028	-0.003	0.009	0.028	0.032	0.030	0.032	0.037	0.034
	(0.090)	(0.069)	(0.057)	(0.098)	(0.075)	(0.063)	(0.098)	(0.075)	(0.063)
Real GDP Growth (QoQ)									
Lower/Same as Previous ($\mathbb{I}_r^- = 1$)	-0.016	0.045	0.065	0.004	0.061	0.046	0.006	0.066	0.051
	(0.097)	(0.072)	(0.061)	(0.104)	(0.078)	(0.065)	(0.104)	(0.078)	(0.065)
Higher than Previous ($\mathbb{I}_r^+ = 1$)	0.147	0.043	0.014	0.185*	0.040	0.038	0.180*	0.038	0.036
	(0.090)	(0.066)	(0.055)	(0.098)	(0.072)	(0.059)	(0.098)	(0.072)	(0.059)
FOMC Meetings/Calls									
Lower/Same as Previous ($\mathbb{I}_r^- = 1$)	0.015	-0.068	-0.051	0.027	-0.067	-0.049	0.025	-0.070	-0.052
	(0.068)	(0.054)	(0.046)	(0.073)	(0.058)	(0.050)	(0.073)	(0.058)	(0.050)
Higher than Previous ($\mathbb{I}_r^+ = 1$)	0.008	0.061	0.132	0.088	0.103	0.170*	0.088	0.103	0.170*
	(0.138)	(0.111)	(0.095)	(0.148)	(0.119)	(0.102)	(0.148)	(0.119)	(0.102)
Observations	198,507	198,507	198,507	165,208	165,208	165,208	165,208	165,208	165,208
Adjusted R^2	0.034	0.034	0.034	0.066	0.066	0.066	0.066	0.066	0.066

Estimates of $(\hat{\alpha}_r^- - \hat{\beta}_r^-)$ and $(\hat{\alpha}_r^+ - \hat{\beta}_r^+)$ from (2): $\pi_j^e = \boldsymbol{x}_j'\boldsymbol{\gamma} + \sum_{r \in R}(\beta_r^+ \mathbb{I}_r^{b+} + \alpha_r^+ \mathbb{I}_r^{a+} + \beta_r^- \mathbb{I}_r^{b-} + \alpha_r^- \mathbb{I}_r^{a-}) + \varepsilon_j$

The analgous full sample results are in Table 3, whereas the sample here is for September 1982 – May 2019. This sample restricition is used as to employ the Thornton (2005) policy rate changes for the September 1982 – July 1990 period, while the remaining policy rate changes are provided directly on the Federal Reverse website.

Notes: $^{***}p < 0.01, ^{**}p < 0.05, ^{*}p < 0.10$. Robust standard errors are in parentheses. Bolded column headings are the control variables included, defined in Table 2, and the columns labeled with $w = 1, 2, 3$ denote the number of days before and after used to define the release window. Dependent variable is individuals reported on year ahead expected inflation rate from Michigan Survey of Consumers (MSC), excluding individuals with reported expected inflation rate greater than 20 percent in absolute value. For releases of the employment situation, consumer price index, industrial production, gross domestic product, and FOMC meetings/calls, the headline series used is in **bold**. The rows labeled *Lower/Same as Previous* denotes instances where the headline series is lower or the same as the previously known value, whereas *Higher than Previous* denotes the converse. These correspond to the negative and positive superscripts, respectively.

do not strongly respond to contractionary nor expansionary policy announcements for the federal funds target rate (*higher than previous, lower than previous, respectively*).

When instead considering the size of changes over this sample, presented in Table 7, the later sample suggests a similar response to the size of changes in the headline unemployment rate, shown in the first row. Additionally, the last row of the table shows that household inflation expectations around FOMC meetings do not significantly adjust to the size of interest rate

Table 7 – Size of Release Changes, September 1982 – May 2019

Dep Var: 1yr Inf. Exp.	**Base Controls**			**+ Sentiment and News**			**+ Macro Aggregates**		
Coef: $\Delta_r^v(\hat{\alpha}_r - \hat{\beta}_r)$	$w=1$	$w=2$	$w=3$	$w=1$	$w=2$	$w=3$	$w=1$	$w=2$	$w=3$
Unemployment Rate ($\Delta_r^v = 1$ p.p.)	-0.701**	-0.581**	-0.355*	-0.688**	-0.594**	-0.456**	-0.693**	-0.604**	-0.466**
	(0.303)	(0.229)	(0.190)	(0.324)	(0.247)	(0.205)	(0.324)	(0.246)	(0.205)
CPI Inflation (MoM, $\Delta_r^v = 1$ a.p.p.)	0.038*	0.026	0.020	0.029	0.006	0.007	0.030	0.006	0.006
	(0.021)	(0.016)	(0.013)	(0.023)	(0.017)	(0.014)	(0.023)	(0.017)	(0.014)
IP Growth (MoM, $\Delta_r^v = 1$ a.p.p.)	-0.011	-0.012*	-0.006	-0.014	-0.014**	-0.007	-0.013	-0.013*	-0.007
	(0.008)	(0.006)	(0.005)	(0.009)	(0.007)	(0.006)	(0.009)	(0.007)	(0.006)
Real GDP Growth (QoQ , $\Delta_r^v = 1$ a.p.p.)	0.025	0.016	0.002	0.039	0.029	0.020	0.037	0.026	0.019
	(0.051)	(0.038)	(0.031)	(0.055)	(0.041)	(0.033)	(0.055)	(0.041)	(0.033)
FOMC Meetings/Calls ($\Delta_r^v = 1$ p.p.)	0.141	0.301	0.445*	0.137	0.191	0.406	0.151	0.211	0.427*
	(0.350)	(0.272)	(0.233)	(0.379)	(0.291)	(0.249)	(0.378)	(0.290)	(0.249)
Observations	198,507	198,507	198,507	165,208	165,208	165,208	165,208	165,208	165,208
Adjusted R^2	0.034	0.034	0.034	0.066	0.066	0.066	0.066	0.066	0.066

Estimates of $\Delta_r^v(\hat{\alpha}_r - \hat{\beta}_r)$ from (3): $\pi_j^e = \boldsymbol{x}_j'\boldsymbol{\gamma} + \sum_{r\in R}\Delta_r^v(\beta_r\mathbb{I}_r^b + \alpha_r\mathbb{I}_r^a) + \varepsilon_j$

The analgous full sample results are in Table 4, whereas the sample here is for September 1982 – May 2019. This sample restricition is used as to employ the Thornton (2005) policy rate changes for the September 1982 – July 1990 period, while the remaining policy rate changes are provided directly on the Federal Reverse website.

Notes: $^{***}p < 0.01, ^{**}p < 0.05, ^{*}p < 0.10$. Robust standard errors are in parentheses. Bolded column headings are the control variables included, defined in Table 2, and the columns labeled with $w = 1, 2, 3$ denote the number of days before and after used to define the release window. Dependent variable is individuals reported on year ahead expected inflation rate from Michigan Survey of Consumers (MSC), excluding individuals with reported expected inflation rate greater than 20 percent in absolute value. For releases of the employment situation, consumer price index, industrial production, gross domestic product, and FOMC meetings/calls, the headline series used is in **bold**. Each row shows the estimated percentage point response of household inflation expectations for each percentage point (p.p.) or annualized percentage point (a.p.p.) change in the corresponding headline series.

changes. Together, even when incorporating additional information from FOMC meetings over a slightly shorter sample, the previous sub-sections results remain unchanged.

5 Does the State of the Economy Matter?

The first portion of this section estimates whether household inflation expectations respond differently to releases that occur during and outside of recessions. The second portion considers an extended window around official recessions.

5.1 NBER Recessions

As I did in Section 4, I slightly modify (1) to assess whether there is a difference in the relevance of the information convey within these releases during and outside of recessions. In a straightforward manner, this simply requires adding an indicator function for whether the release occurred during an officially declared recession or not, yielding:

$$\pi_j^e = \boldsymbol{x}_j'\boldsymbol{\gamma} + \sum_{r\in R}\mathbb{I}_r^{nber}\left(\beta_r\mathbb{I}_r^b + \alpha_r\mathbb{I}_r^a\right) + \varepsilon_j \tag{5}$$

$$\pi_j^e = \boldsymbol{x}_j'\boldsymbol{\gamma} + \sum_{r\in R}\mathbb{I}_r^{nber}\Delta_r^v\left(\beta_r\mathbb{I}_r^b + \alpha_r\mathbb{I}_r^a\right) + \varepsilon_j \tag{6}$$

Where $\mathbb{I}_r^{nber} = 1$ if release occurred during recession, and is zero otherwise. For convenience, I have presented these specifications utilizing this simple indicator, however in estimation it should be noted that two sets of coefficients are being estimated, one for releases during recessions and another for releases occurring outside recessions. Here, using the survey month relative to the interview date makes no difference because the highest frequency for official recessions is monthly.

While (6) is the same *size* of changes equation as (3), simply outfitted with the $\mathbb{I}_r^{nber}$ indicator, (5) is *not* explicitly estimating the response of household inflation expectations to the *sign* of release changes. This is because the overwhelming majority of these changes in each releases headline series during recessions are in one direction, typically indicating an increasing unemployment rate and decreases in the remaining releases. As such, one cannot separately identify responses around both *sign* types considered previously when those releases occur during recessions. However, in light of the observation that these changes typically indicate a particular direction of change in each headline series, (5) amounts to modifying (1) to incorporating the recession indicator. In turn, I argue that this is capturing the intended response, at least during recession releases of each series.

Further, in light of the negligible difference in results between the full sample and sub-sample results in Section 4, I only present estimation results for the full sample in this section. This means that for FOMC meetings/calls, there is no *size* incorporated into the regression. Additionally, if one were to focus on the later sub-sample starting in September 1982, the truncation would omit the entirety of releases and interviews conducted in the February 1980 – July 1980 recession, as well as most of the observations in the August 1981 – November 1982 recession.[29] Thus, estimation would only include the recessions occurring between August 1990 – March 1991, April 2001 – November 2001, and January 2008 – June 2009.

The importance of release of the unemployment rate are reflected outside and during recessions as well. When not incorporating the change in headline series, the estimates in Table 8 suggest that household inflation expectations are about 0.3 percentage points lower after unemployment releases relative to before. Instead of simple before and after indicators, Table 9 estimates the response of inflation expectations with the size of each release change. The magnitude of the coefficient estimates shown in the first row under the unemployment rate grouping are quite similar to the estimates found in Table 4. In other words, household inflation expectations respond strongly to the size of changes in the unemployment rate outside of

29: Discussed further in the next section, Table 10 contains the start and end dates of NBER recessions over the full sample.

Table 8 – Are Inflation Expectations Different After Releases that Occur During Recessions?

Dep Var: 1yr Inf. Exp.	**Base Controls**			**+ Sentiment and News**			**+ Macro Aggregates**		
Coef: $\mathbb{I}_r^{nber}(\hat{\alpha}_r - \hat{\beta}_r)$	$w = 1$	$w = 2$	$w = 3$	$w = 1$	$w = 2$	$w = 3$	$w = 1$	$w = 2$	$w = 3$
Employment Situatuion									
Outside Recessions ($\mathbb{I}_r^{nber} = 0$)	-0.021	-0.023	-0.008	0.012	0.010	0.033	0.013	0.009	0.032
	(0.049)	(0.038)	(0.032)	(0.053)	(0.041)	(0.034)	(0.053)	(0.041)	(0.034)
During Recessions ($\mathbb{I}_r^{nber} = 1$)	-0.337*	-0.305**	-0.197*	-0.339*	-0.369**	-0.219*	-0.345*	-0.382**	-0.240**
	(0.187)	(0.139)	(0.114)	(0.201)	(0.149)	(0.122)	(0.201)	(0.148)	(0.122)
Consumer Price Index									
Outside Recessions ($\mathbb{I}_r^{nber} = 0$)	-0.114*	-0.017	-0.070*	-0.121*	-0.032	-0.088**	-0.121*	-0.032	-0.089**
	(0.064)	(0.049)	(0.042)	(0.069)	(0.052)	(0.045)	(0.069)	(0.052)	(0.045)
During Recessions ($\mathbb{I}_r^{nber} = 1$)	0.290	0.148	0.190	0.167	0.100	0.141	0.196	0.132	0.183
	(0.264)	(0.201)	(0.170)	(0.292)	(0.220)	(0.186)	(0.291)	(0.220)	(0.185)
Industrial Production									
Outside Recessions ($\mathbb{I}_r^{nber} = 0$)	0.033	0.009	0.001	0.074	0.064	0.034	0.077	0.066	0.037
	(0.063)	(0.048)	(0.041)	(0.068)	(0.052)	(0.044)	(0.068)	(0.052)	(0.044)
During Recessions ($\mathbb{I}_r^{nber} = 1$)	0.156	-0.035	0.025	0.207	-0.001	0.075	0.195	-0.032	0.019
	(0.251)	(0.194)	(0.165)	(0.271)	(0.208)	(0.178)	(0.270)	(0.208)	(0.178)
Gross Domestic Product									
Outside Recessions ($\mathbb{I}_r^{nber} = 0$)	0.084	0.059	0.057	0.113	0.049	0.046	0.114	0.053	0.051
	(0.067)	(0.050)	(0.041)	(0.072)	(0.053)	(0.044)	(0.072)	(0.053)	(0.044)
During Recessions ($\mathbb{I}_r^{nber} = 1$)	0.109	0.057	0.016	0.033	0.016	0.014	0.047	0.021	0.012
	(0.251)	(0.190)	(0.158)	(0.276)	(0.205)	(0.170)	(0.275)	(0.204)	(0.170)
FOMC Meetings/Calls									
Outside Recessions ($\mathbb{I}_r^{nber} = 0$)	0.037	-0.009	0.026	0.062	0.002	0.040	0.068	0.007	0.043
	(0.062)	(0.050)	(0.043)	(0.067)	(0.054)	(0.046)	(0.067)	(0.054)	(0.046)
During Recessions ($\mathbb{I}_r^{nber} = 1$)	-0.191	-0.060	-0.072	-0.253	-0.093	-0.104	-0.272	-0.100	-0.096
	(0.195)	(0.149)	(0.128)	(0.208)	(0.159)	(0.138)	(0.207)	(0.159)	(0.138)
Observations	221,845	221,845	221,845	184,930	184,930	184,930	184,930	184,930	184,930
Adjusted R^2	0.086	0.086	0.086	0.119	0.119	0.119	0.119	0.119	0.119

Estimates of $\mathbb{I}_r^{nber}(\hat{\alpha}_r - \hat{\beta}_r)$ outside and during recessions from (5): $\pi_j^e = \boldsymbol{x}_j'\boldsymbol{\gamma} + \sum_{r \in R} \mathbb{I}_r^{nber}(\beta_r \mathbb{I}_r^b + \alpha_r \mathbb{I}_r^a) + \varepsilon_j$

Notes: $^{***}p < 0.01, ^{**}p < 0.05, ^{*}p < 0.10$. Robust standard errors are in parentheses. Sample is for June 1979 – May 2019. Bolded column headings are the control variables included, defined in Table 2, and the columns labeled with $w = 1, 2, 3$ denote the number of days before and after used to define the release window. Dependent variable is individuals reported on year ahead expected inflation rate from Michigan Survey of Consumers (MSC), excluding individuals with reported expected inflation rate greater than 20 percent in absolute value. For releases of the employment situation, consumer price index, industrial production, gross domestic product, and FOMC meetings/calls, the headline series used is in **bold**. The rows labeled *Outside Recessions* denotes releases that did not occur during months that NBER classified as recessions, whereas *During Recessions* denotes estimates for releases that did occur during these months.

recessions.

5.2 Extending the Definition of Recessions

Given that interview dates do not always coincide with the survey month in which each respondents answers may appear, it seems odd that, absent some macroeconomic release, that a person interviewed in December 2007 could not reasonably be considered as living in a recessionary-like state. To understand the dynamics of inflation expectations preceding and following recessions, I modify (6) to include additional indicators for releases occurring within

Table 9 – Does the Size of Change in each Headline Series During Recessions Matter?

Dep Var: 1yr Inf. Exp.	**Base Controls**			**+ Sentiment and News**			**+ Macro Aggregates**		
Coef: $\mathbb{I}_r^{nber}\Delta_r^v(\hat{\alpha}_r - \hat{\beta}_r)$	$w=1$	$w=2$	$w=3$	$w=1$	$w=2$	$w=3$	$w=1$	$w=2$	$w=3$
Unemployment Rate ($\Delta_r^v = 1$ p.p.)									
Outside Recessions ($\mathbb{I}_r^{nber}=0$)	-0.738**	-0.555**	-0.353*	-0.718**	-0.502**	-0.429**	-0.731**	-0.505**	-0.430**
	(0.313)	(0.238)	(0.198)	(0.339)	(0.256)	(0.213)	(0.338)	(0.256)	(0.213)
During Recessions ($\mathbb{I}_r^{nber}=1$)	-0.444	-0.578	-0.121	-0.582	-0.756	-0.187	-0.585	-0.774	-0.230
	(0.632)	(0.458)	(0.376)	(0.671)	(0.486)	(0.399)	(0.670)	(0.486)	(0.399)
CPI Inflation (MoM, $\Delta_r^v = 1$ a.p.p.)									
Outside Recessions ($\mathbb{I}_r^{nber}=0$)	0.019	-0.001	-0.004	0.003	-0.025	-0.022	0.004	-0.024	-0.021
	(0.020)	(0.015)	(0.013)	(0.021)	(0.016)	(0.014)	(0.021)	(0.016)	(0.014)
During Recessions ($\mathbb{I}_r^{nber}=1$)	0.068	0.083**	0.053	0.080	0.072*	0.046	0.082	0.075*	0.050
	(0.053)	(0.039)	(0.034)	(0.058)	(0.043)	(0.036)	(0.057)	(0.043)	(0.036)
IP Growth (MoM, $\Delta_r^v = 1$ a.p.p.)									
Outside Recessions ($\mathbb{I}_r^{nber}=0$)	0.001	-0.003	0.002	0.000	-0.005	0.003	0.000	-0.005	0.003
	(0.008)	(0.006)	(0.005)	(0.008)	(0.006)	(0.005)	(0.008)	(0.006)	(0.005)
During Recessions ($\mathbb{I}_r^{nber}=1$)	-0.046**	-0.021	-0.019	-0.048**	-0.017	-0.018	-0.047**	-0.016	-0.016
	(0.021)	(0.017)	(0.014)	(0.022)	(0.018)	(0.015)	(0.022)	(0.018)	(0.015)
Real GDP Growth (QoQ, $\Delta_r^v = 1$ a.p.p.)									
Outside Recessions ($\mathbb{I}_r^{nber}=0$)	0.037	0.028	0.017	0.043	0.031	0.022	0.043	0.031	0.023
	(0.053)	(0.038)	(0.031)	(0.057)	(0.041)	(0.033)	(0.057)	(0.041)	(0.033)
During Recessions ($\mathbb{I}_r^{nber}=1$)	0.019	0.072	0.048	0.075	0.109	0.072	0.072	0.108	0.077
	(0.114)	(0.079)	(0.064)	(0.125)	(0.084)	(0.066)	(0.125)	(0.084)	(0.066)
FOMC Meetings/Calls (No Size)									
Outside Recessions ($\mathbb{I}_r^{nber}=0$)	0.042	0.011	0.034	0.069	0.026	0.044	0.072	0.027	0.046
	(0.062)	(0.049)	(0.042)	(0.067)	(0.053)	(0.046)	(0.067)	(0.053)	(0.046)
During Recessions ($\mathbb{I}_r^{nber}=1$)	-0.218	-0.069	-0.048	-0.300	-0.111	-0.082	-0.291	-0.102	-0.074
	(0.194)	(0.148)	(0.128)	(0.207)	(0.159)	(0.138)	(0.207)	(0.158)	(0.138)
Observations	221,845	221,845	221,845	184,930	184,930	184,930	184,930	184,930	184,930
Adjusted R^2	0.087	0.087	0.087	0.119	0.119	0.119	0.120	0.120	0.120

Estimates of $\mathbb{I}_r^{nber}\Delta_r^v(\hat{\alpha}_r - \hat{\beta}_r)$ from (6): $\pi_j^e = \boldsymbol{x}_j'\boldsymbol{\gamma} + \sum_{r\in R}\mathbb{I}_r^{nber}\Delta_r^v(\beta_r\mathbb{I}_r^b + \alpha_r\mathbb{I}_r^a) + \varepsilon_j$

Notes: $^{***}p<0.01, ^{**}p<0.05, ^{*}p<0.10$. Robust standard errors are in parentheses. Sample is for June 1979 – May 2019. Bolded column headings are the control variables included, defined in Table 2, and the columns labeled with $w = 1, 2, 3$ denote the number of days before and after used to define the release window. Dependent variable is individuals reported on year ahead expected inflation rate from Michigan Survey of Consumers (MSC), excluding individuals with reported expected inflation rate greater than 20 percent in absolute value. For releases of the employment situation, consumer price index, industrial production, gross domestic product, and FOMC meetings/calls, the headline series used is in **bold**. The rows labeled *Outside Recessions* and *During Recessions* show the estimated percentage point response in household inflation expectations to a percentage point (p.p.) or annualized percentage point (a.p.p) change in the headline series depending on whether the release occurred during a NBER designated recession or outside of one.

the 6 months preceding any recession, and another for releases in the 6 months following a recession. Further, just like the previous portion of this section, releases occurring during the official recession dates also receive their own indicator.

Altogether, this will additionally capture the dynamics of household inflation expectations during this specific portion of historical business cycles. In the first column of Table 10, the official start and end dates of NBER recessions over the sample period are shown. For each of these, the second and third columns of the table show the corresponding 6 months preceding and following the recession. Releases occurring during these periods will receive the aforementioned

Table 10 – Extended Definition of Recessions

NBER Recessions	Preceding 6 Months	Following 6 Months
Feb 1980 – Jul 1980	Aug 1979 – Jan 1980	Aug 1980 – Jan 1981
Aug 1981 – Nov 1982	Feb 1981 – Jul 1981	Dec 1982 – May 1983
Aug 1990 – Mar 1991	Feb 1990 – Jul 1990	Apr 1991 – Sep 1991
Apr 2001 – Nov 2001	Oct 2000 – Mar 2001	Dec 2001 – May 2002
Jan 2008 – Jun 2009	Jul 2007 – Dec 2007	Jul 2009 – Dec 2009

Notes: These are the extended recession window dates used in the regression results presented in **??** and Table 11. The first column are the official start and end dates of NBER recessions occurring between June 1979 and May 2019. The second and third columns are the time periods for which releases will get an additional indicators.

indicator. Table 11 shows estimates of (6) with extended recession indicators. The columns remain the same as previously discussed, and under each release type the coefficient estimate for the adjustment in household inflation expectations are shown for releases within 6 months before or after the official start or end of an recessions (*preceding 6 months, following 6 months, respectively*). Relative to previous results, one immediately recognizes a number of additional insights.

First, household inflation expectations respond quite strongly, both in magnitude and statistical significance, to changes in the unemployment rate in the initial months of recovery after a recession. For the tightest identification window, $w = 1$, a 1 percentage point decrease in the unemployment rate, the typical direction of these changes following recessions, suggests household inflation expectations increase by about 3.5 percentage points. Here, this suggests that as the economy begins recovery, households are keenly aware of labor market conditions as they begin increasing their inflation expectations. This is further supported by referring back to Figure 1, which showed the daily series of household inflation expectations and the vintage releases values, aligned to when the release historically occurred.

While much smaller in magnitude, this increase in inflation expectations is potentially somewhat offset by the increasing inflation rate, highlighted by the significant response when using the 2 and 3 day identification windows under CPI inflation. While very large swings in the month-over-month headline inflation rate do occur, the approximate 0.15 percentage point response to a 1 annualized percentage point change is about 23 times smaller than how households adjust in light of information about the unemployment rate. Further, the last row under FOMC meetings shows some suggestive evidence that monetary policy may serve a role in adjusting household inflation expectations in these initial months. Although these results do not include intended policy rate changes, as previously discussed, these 6 months following recessions correspond directly to either maintaining the recession easing of policy or further

Table 11 – Does the Size of the Change in each Headline Series Matter Around Recessions?

Dep Var: 1yr Inf. Exp.	**Base Controls**			**+ Sentiment and News**			**+ Macro Aggregates**		
Coef: $\mathbb{I}_r^{ext}\Delta_r^v(\hat{\alpha}_r - \hat{\beta}_r)$	$w=1$	$w=2$	$w=3$	$w=1$	$w=2$	$w=3$	$w=1$	$w=2$	$w=3$
Unemployment Rate ($\Delta_r^v = 1$ p.p.)									
Preceding 6 Months	-0.626	-0.765	-0.636	-0.571	-0.293	0.091	-0.625	-0.323	0.061
	(1.501)	(1.124)	(0.917)	(1.599)	(1.176)	(0.970)	(1.599)	(1.176)	(0.970)
During Recessions	-0.442	-0.574	-0.117	-0.581	-0.753	-0.185	-0.584	-0.770	-0.228
	(0.632)	(0.458)	(0.376)	(0.671)	(0.486)	(0.399)	(0.670)	(0.486)	(0.399)
Following 6 Months	-3.484***	-1.914**	-0.847	-3.551***	-2.098**	-1.003	-3.642***	-2.145**	-1.039
	(1.184)	(0.864)	(0.700)	(1.269)	(0.902)	(0.744)	(1.268)	(0.901)	(0.744)
CPI Inflation (MoM, $\Delta_r^v = 1$ a.p.p.)									
Preceding 6 Months	-0.063	-0.043	-0.062	-0.020	-0.068	-0.068	-0.020	-0.068	-0.068
	(0.071)	(0.054)	(0.047)	(0.072)	(0.058)	(0.050)	(0.072)	(0.058)	(0.051)
During Recessions	0.068	0.083**	0.053	0.080	0.072*	0.046	0.083	0.075*	0.050
	(0.053)	(0.039)	(0.034)	(0.058)	(0.043)	(0.036)	(0.057)	(0.043)	(0.036)
Following 6 Months	0.027	-0.082	-0.066	-0.010	-0.144**	-0.108**	-0.003	-0.137**	-0.103*
	(0.079)	(0.060)	(0.050)	(0.084)	(0.064)	(0.053)	(0.084)	(0.064)	(0.053)
IP Growth (MoM, $\Delta_r^v = 1$ a.p.p.)									
Preceding 6 Months	-0.012	0.001	-0.004	0.012	0.010	0.000	0.013	0.011	0.001
	(0.029)	(0.023)	(0.020)	(0.031)	(0.025)	(0.021)	(0.031)	(0.025)	(0.021)
During Recessions	-0.047**	-0.022	-0.019	-0.048**	-0.017	-0.018	-0.048**	-0.016	-0.016
	(0.021)	(0.017)	(0.014)	(0.022)	(0.018)	(0.015)	(0.022)	(0.018)	(0.015)
Following 6 Months	0.044	-0.007	0.000	0.044	-0.005	0.008	0.043	-0.006	0.007
	(0.041)	(0.029)	(0.023)	(0.045)	(0.032)	(0.025)	(0.045)	(0.032)	(0.025)
Real GDP Growth (QoQ, $\Delta_r^v = 1$ a.p.p.)									
Preceding 6 Months	-0.101	-0.056	-0.068	0.038	0.096	0.088	0.024	0.081	0.074
	(0.169)	(0.122)	(0.103)	(0.189)	(0.138)	(0.114)	(0.189)	(0.137)	(0.114)
During Recessions	0.019	0.072	0.047	0.075	0.108	0.072	0.072	0.108	0.076
	(0.114)	(0.079)	(0.064)	(0.125)	(0.084)	(0.066)	(0.125)	(0.084)	(0.066)
Following 6 Months	0.160	0.104	0.139*	0.142	0.069	0.093	0.146	0.072	0.098
	(0.124)	(0.096)	(0.072)	(0.128)	(0.103)	(0.077)	(0.128)	(0.103)	(0.077)
FOMC Meetings/Calls (No Size)									
Preceding 6 Months	0.073	-0.031	-0.002	0.041	-0.146	-0.042	0.032	-0.154	-0.051
	(0.243)	(0.188)	(0.162)	(0.254)	(0.200)	(0.172)	(0.254)	(0.200)	(0.172)
During Recessions	-0.219	-0.069	-0.048	-0.300	-0.110	-0.082	-0.291	-0.100	-0.073
	(0.194)	(0.148)	(0.128)	(0.207)	(0.159)	(0.138)	(0.207)	(0.158)	(0.138)
Following 6 Months	0.149	0.242	0.161	0.289	0.432**	0.287*	0.312	0.449**	0.313*
	(0.244)	(0.186)	(0.157)	(0.269)	(0.201)	(0.170)	(0.268)	(0.201)	(0.170)
Observations	221,845	221,845	221,845	184,930	184,930	184,930	184,930	184,930	184,930
Adjusted R^2	0.087	0.087	0.087	0.120	0.120	0.120	0.120	0.120	0.120

Estimates of $\mathbb{I}_r^{ext}\Delta_r^v(\hat{\alpha}_r - \hat{\beta}_r)$ from (6): $\pi_j^e = \boldsymbol{x}_j'\boldsymbol{\gamma} + \sum_{r\in R}\mathbb{I}_r^{ext}\Delta_r^v(\beta_r\mathbb{I}_r^b + \alpha_r\mathbb{I}_r^a) + \varepsilon_j$

Notes: $^{***}p<0.01, ^{**}p<0.05, ^{*}p<0.10$. Robust standard errors are in parentheses. Sample is for June 1979 – May 2019. Bolded column headings are the control variables included, defined in Table 2, and the columns labeled with $w = 1, 2, 3$ denote the number of days before and after used to define the release window. Dependent variable is individuals reported on year ahead expected inflation rate from Michigan Survey of Consumers (MSC), excluding individuals with reported expected inflation rate greater than 20 percent in absolute value. For releases of the employment situation, consumer price index, industrial production, gross domestic product, and FOMC meetings/calls, the headline series used is in **bold**. Extended recessions refers to adding additional indicators for instances of releases occurring in the preceding 6 months before the offical NBER recession start month, as well as an additional indicator for the following 6 months after their end. These dates are provided in Table 10. Estimates for releases outside recessions are omitted, but provided in Table A.3. Each row shows the estimated percentage point response of household inflation expectations to a percentage point (p.p.) or annualized percentage point (a.p.p.) change in the headline series for releases occurring within the time period denoted in *italics*.

easing.

6 Anticipated v. Unanticipated Changes

To conclude my results, I consider how household inflation expectations respond to the unanticipated information conveyed by each release. To this end, I measure the unanticipated portion by incorporating the consensus forecasts of each release from Bloomberg Survey. This includes various professional forecasters, ranging from commercial banks to market experts, with forecasts available starting in November 1996. For each release, forecasters submit their individual expectation for the headline estimate, thus allowing one to gauge how each release deviated from their expectations. Similarly to previous results, I consider the*size* of the information conveyed by each release, however now these will be relative to the *expectations* of the Bloomberg forecasters.[30] For clarity, the *size* of the unanticipated information conveyed by the release would be the difference between the observed headline series and the consensus forecast from before the release occurred. To remain consistent with the literature, I will refer to this as the *surprise*, and estimation only requires only minor modification of (3):

$$\pi_j^e = \boldsymbol{x}_j'\boldsymbol{\gamma} + \sum_{r \in R} fe_r \left(\beta_r \mathbb{I}_r^b + \alpha_r \mathbb{I}_r^a\right) + \varepsilon_j \tag{7}$$

The interview window indicators remain unchanged from their previous definition, but now the *size* of this surprise is included in (7) as $fe_r = v_r^{new} - v_r^e$, where the latter is the consensus expectation discussed previously. Additionally, this estimates the percentage point response in household inflation expectations for every p.p. or a.p.p. increase in the *size* of the surprise.[31] Table 12 shows estimates of (7) for November 1996 – May 2019.

Interestingly, this later sample suggests that household inflation expectations respond to unexpected deviations in both IP growth and Real GDP growth. For example, when the release of the previous quarters Real GDP growth rate is 1 annualized percentage point higher than expected by forecasters, household inflation expectations are revised downward by about 0.5 percentage points. This inverse relationship with new output information would be inconsistent with the sign of the estimates if households believed, to some degree, that the higher than expected growth was attributable to demand factors. Instead, this suggests that households generally view these surprises as being drive by supply-side factors, and that they view these factors as having an effect over at least the next year.[32]

30: I also considered the sign of the surprises, but my estimates suggested that households do not adjustment inflation expectations in response to any of the releases in that case.

31: While not discussed explicitly within the main text, Table A.4 provides estimates for the size of the change in the headline series as well.

32: This follows from the dependent variable being their one-year ahead expected inflation rate.

Table 12 – Size of Surprises in each Headline Series, Nov 1996 – May 2019

Dep Var: 1yr Inf. Exp.	**Base Controls**			**+ Sentiment and News**			**+ Macro Aggregates**		
Coef: $fe_r(\hat{\alpha}_r - \hat{\beta}_r)$	$w=1$	$w=2$	$w=3$	$w=1$	$w=2$	$w=3$	$w=1$	$w=2$	$w=3$
Unemployment Rate (fe_r = 1 p.p.)	-0.726	-0.102	-0.086	-0.472	0.129	-0.010	-0.454	0.132	0.004
	(0.443)	(0.335)	(0.278)	(0.465)	(0.353)	(0.294)	(0.464)	(0.352)	(0.293)
CPI Inflation (MoM, fe_r = 1 a.p.p.)	0.055	0.004	0.017	0.046	-0.027	0.014	0.049	-0.025	0.014
	(0.059)	(0.044)	(0.037)	(0.063)	(0.047)	(0.038)	(0.063)	(0.046)	(0.038)
IP Growth (MoM, fe_r = 1 a.p.p.)	-0.034	-0.038**	-0.031**	-0.032	-0.039**	-0.026*	-0.032	-0.039**	-0.025*
	(0.022)	(0.016)	(0.014)	(0.023)	(0.017)	(0.014)	(0.023)	(0.017)	(0.014)
Real GDP Growth (QoQ , fe_r = 1 a.p.p.)	-0.443**	-0.208	-0.116	-0.520**	-0.201	-0.034	-0.529***	-0.199	-0.031
	(0.192)	(0.140)	(0.114)	(0.204)	(0.149)	(0.121)	(0.205)	(0.149)	(0.121)
FOMC Meetings/Calls (fe_r = 1 p.p.)	-0.397	-0.189	-0.423	-1.199	-1.609	-1.244	-1.212	-1.600	-1.244
	(2.029)	(1.861)	(1.530)	(2.261)	(1.940)	(1.615)	(2.234)	(1.915)	(1.594)
Observations	115,979	115,979	115,979	98,507	98,507	98,507	98,507	98,507	98,507
Adjusted R^2	0.044	0.044	0.044	0.087	0.087	0.087	0.088	0.088	0.088

Estimates of $fe_r(\hat{\alpha}_r - \hat{\beta}_r)$ from (7): $\pi_j^e = \boldsymbol{x}_j'\boldsymbol{\gamma} + \sum_{r\in R} fe_r(\beta_r \mathbb{I}_r^b + \alpha_r \mathbb{I}_r^a) + \varepsilon_j$

Notes: $^{***}p < 0.01, ^{**}p < 0.05, ^{*}p < 0.10$. Robust standard errors are in parentheses. Sample is for November 1996 – May 2019. Bolded column headings are the control variables included, defined in Table 2, and the columns labeled with $w = 1, 2, 3$ denote the number of days before and after used to define the release window. Dependent variable is individuals reported on year ahead expected inflation rate from Michigan Survey of Consumers (MSC), excluding individuals with reported expected inflation rate greater than 20 percent in absolute value. For releases of the employment situation, consumer price index, industrial production, gross domestic product, and FOMC meetings/calls, the headline series used is in **bold**. Each row shows the estimated percentage point response of household inflation expectations for each percentage point (p.p.) or annualized percentage point (a.p.p.) difference between the headline series release and the consensus forecast from Bloomberg. For comparsion, Table A.4 provides estimates for how household inflation expectations respond to the size of the change in the headline series relative to its *previously known value* for the shorter time sample.

7 Conclusion

This paper exploits an quasi-experimental setting to investigate whether the inflation expectations from the Michigan Survey of Consumers (MSC) have historically responded to macroeconomic data releases. This setting is novel, stemming from the recording of interview dates for each respondent starting in June 1979, and publicly accessible for over 250,000 MSC participants until May 2019. Leveraging this, the study sidesteps previous identification challenges by pinpointing interviews occurring within one to three days of five distinct macroeconomic releases. This setup enables estimating the impact of these releases on household inflation expectations individually, disentangling the effects from other releases. The findings uncover a responsive pattern in household inflation expectations tied to certain macroeconomic data.

Notably, insights regarding the unemployment rate and CPI inflation wield significant influence. Interestingly, expectations are more sensitive to unemployment rate increases than decreases. Similarly, shifts in headline CPI inflation, particularly when lower than previously reported, prompt adjustments in expectations. An intriguing asymmetry surfaces surrounding recession periods, showcasing robust reactions in inflation expectations for six months following such economic downturns. Overall, these findings underscore the adaptability of household inflation expectations in response to macroeconomic releases, emphasizing the varied impact

of different information. The focal releases, unemployment rate, and CPI inflation, align with the Federal Reserve's dual mandate, suggesting the potential for central bank interventions to shape household inflation expectations within its mandated policies.

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Appendix A – Additional Results and Robustness

A.1: Additional Results for Baseline/FOMC

Table A.1 – Estimates for Equation (1) in Section 4.1

Dep Var: 1yr Inf. Exp.	**Base Controls**			**+ Sentiment and News**			**+ Macro Aggregates**		
Coef: $(\hat{\alpha}_r - \hat{\beta}_r)$	$w=1$	$w=2$	$w=3$	$w=1$	$w=2$	$w=3$	$w=1$	$w=2$	$w=3$
Employment Situation	-0.053	-0.012	-0.005	-0.018	0.011	0.024	-0.017	0.011	0.025
	(0.033)	(0.028)	(0.024)	(0.035)	(0.030)	(0.026)	(0.035)	(0.030)	(0.026)
Consumer Price Index	-0.004	0.011	-0.022	-0.011	-0.003	-0.040	-0.010	-0.002	-0.040
	(0.041)	(0.036)	(0.032)	(0.045)	(0.039)	(0.035)	(0.045)	(0.039)	(0.035)
Industrial Production	-0.068*	-0.068*	-0.056*	-0.030	-0.024	-0.014	-0.032	-0.027	-0.017
	(0.041)	(0.035)	(0.032)	(0.044)	(0.038)	(0.034)	(0.044)	(0.038)	(0.034)
Gross Domestic Product	0.097**	0.071**	0.055**	0.105**	0.083**	0.057*	0.106**	0.084**	0.058*
	(0.039)	(0.032)	(0.028)	(0.042)	(0.035)	(0.030)	(0.042)	(0.035)	(0.030)
FOMC Meetings/Calls	0.027	0.021	0.032	-0.002	-0.002	0.012	0.002	0.001	0.015
	(0.035)	(0.031)	(0.028)	(0.038)	(0.033)	(0.030)	(0.038)	(0.033)	(0.030)
Observations	221,845	221,845	221,845	184,930	184,930	184,930	184,930	184,930	184,930
Adjusted R^2	0.086	0.086	0.086	0.119	0.119	0.119	0.119	0.119	0.119

Estimates of $(\hat{\alpha}_r - \hat{\beta}_r)$ from (1): $\pi_j^e = \boldsymbol{x}_j'\boldsymbol{\gamma} + \sum_{r\in R}(\beta_r \mathbb{I}_r^b + \alpha_r \mathbb{I}_r^a) + \varepsilon_j$

Notes: $^{***}p < 0.01, ^{**}p < 0.05, ^{*}p < 0.10$. Robust standard errors are in parentheses. Sample is for June 1979 – May 2019. Bolded column headings are the control variables included, defined in Table 2, and the columns labeled with $w = 1, 2, 3$ denote the number of days before and after used to define the release window. Dependent variable is individuals reported on year ahead expected inflation rate from Michigan Survey of Consumers (MSC), excluding individuals with reported expected inflation rate greater than 20 percent in absolute value. For releases of the employment situation, consumer price index, industrial production, gross domestic product, and FOMC meetings/calls, the headline series used is in **bold**.

Table A.2 – Estimates for Equation (1) in Section 4.2

Dep Var: 1yr Inf. Exp.	**Base Controls**			**+ Sentiment and News**			**+ Macro Aggregates**		
Coef: $(\hat{\alpha}_r - \hat{\beta}_r)$	$w=1$	$w=2$	$w=3$	$w=1$	$w=2$	$w=3$	$w=1$	$w=2$	$w=3$
Employment Situation	-0.044	-0.029	-0.018	-0.021	-0.009	0.012	-0.020	-0.008	0.013
	(0.033)	(0.028)	(0.024)	(0.035)	(0.030)	(0.026)	(0.035)	(0.030)	(0.026)
Consumer Price Index	0.044	0.028	-0.024	0.046	0.022	-0.032	0.045	0.020	-0.034
	(0.042)	(0.036)	(0.032)	(0.045)	(0.039)	(0.035)	(0.045)	(0.039)	(0.035)
Industrial Production	-0.091**	-0.067*	-0.041	-0.071	-0.038	-0.014	-0.072	-0.039	-0.016
	(0.041)	(0.035)	(0.032)	(0.044)	(0.038)	(0.035)	(0.044)	(0.038)	(0.035)
Gross Domestic Product	0.083**	0.063**	0.051*	0.101**	0.081**	0.057*	0.100**	0.082**	0.058*
	(0.039)	(0.032)	(0.028)	(0.042)	(0.034)	(0.030)	(0.042)	(0.034)	(0.030)
FOMC Meetings/Calls	0.011	0.003	0.000	-0.017	-0.017	-0.020	-0.011	-0.012	-0.016
	(0.036)	(0.031)	(0.028)	(0.038)	(0.033)	(0.031)	(0.038)	(0.033)	(0.031)
Observations	198,507	198,507	198,507	165,208	165,208	165,208	165,208	165,208	165,208
Adjusted R^2	0.034	0.033	0.033	0.066	0.066	0.066	0.066	0.066	0.066

Estimates of $(\hat{\alpha}_r - \hat{\beta}_r)$ from (1): $\pi_j^e = \boldsymbol{x}_j'\boldsymbol{\gamma} + \sum_{r\in R}(\beta_r \mathbb{I}_r^b + \alpha_r \mathbb{I}_r^a) + \varepsilon_j$

The full sample results are in Table A.1, whereas the sample here is for September 1982 – May 2019. This sample restricition is used as to employ the Thornton (2005) policy rate changes for the September 1982 – July 1990 period, while the remaining policy rate changes are provided directly on the Federal Reverse website.

Notes: $^{***}p < 0.01, ^{**}p < 0.05, ^{*}p < 0.10$. Robust standard errors are in parentheses. Bolded column headings are the control variables included, defined in Table 2, and the columns labeled with $w = 1, 2, 3$ denote the number of days before and after used to define the release window. Dependent variable is individuals reported on year ahead expected inflation rate from Michigan Survey of Consumers (MSC), excluding individuals with reported expected inflation rate greater than 20 percent in absolute value. For releases of the employment situation, consumer price index, industrial production, gross domestic product, and FOMC meetings/calls, the headline series used is in **bold**.

A.2: Additional Results for Recessions

Table A.3 – Extended Recessions with Size of Change in Headline Series (Table 11)

Dep Var: 1yr Inf. Exp.	**Base Controls**			**+ Sentiment and News**			**+ Macro Aggregates**		
Coef: $\mathbb{I}_r^{ext}\Delta_r^v(\hat{\alpha}_r - \hat{\beta}_r)$	$w=1$	$w=2$	$w=3$	$w=1$	$w=2$	$w=3$	$w=1$	$w=2$	$w=3$
Unemployment Rate (Δ_r^v = 1 p.p.)									
Outside	-0.376	-0.346	-0.246	-0.332	-0.299	-0.379*	-0.337	-0.298	-0.379*
	(0.321)	(0.248)	(0.207)	(0.346)	(0.267)	(0.224)	(0.346)	(0.267)	(0.224)
Preceding 6 Months	-0.626	-0.765	-0.636	-0.571	-0.293	0.091	-0.625	-0.323	0.061
	(1.501)	(1.124)	(0.917)	(1.599)	(1.176)	(0.970)	(1.599)	(1.176)	(0.970)
During Recessions	-0.442	-0.574	-0.117	-0.581	-0.753	-0.185	-0.584	-0.770	-0.228
	(0.632)	(0.458)	(0.376)	(0.671)	(0.486)	(0.399)	(0.670)	(0.486)	(0.399)
Following 6 Months	-3.484***	-1.914**	-0.847	-3.551***	-2.098**	-1.003	-3.642***	-2.145**	-1.039
	(1.184)	(0.864)	(0.700)	(1.269)	(0.902)	(0.744)	(1.268)	(0.901)	(0.744)
CPI Inflation (MoM, Δ_r^v = 1 a.p.p.)									
Outside	0.027	0.015	0.011	0.006	-0.005	-0.004	0.007	-0.005	-0.004
	(0.021)	(0.016)	(0.013)	(0.023)	(0.017)	(0.014)	(0.023)	(0.017)	(0.014)
Preceding 6 Months	-0.063	-0.043	-0.062	-0.020	-0.068	-0.068	-0.020	-0.068	-0.068
	(0.071)	(0.054)	(0.047)	(0.072)	(0.058)	(0.050)	(0.072)	(0.058)	(0.051)
During Recessions	0.068	0.083**	0.053	0.080	0.072*	0.046	0.083	0.075*	0.050
	(0.053)	(0.039)	(0.034)	(0.058)	(0.043)	(0.036)	(0.057)	(0.043)	(0.036)
Following 6 Months	0.027	-0.082	-0.066	-0.010	-0.144**	-0.108**	-0.003	-0.137**	-0.103*
	(0.079)	(0.060)	(0.050)	(0.084)	(0.064)	(0.053)	(0.084)	(0.064)	(0.053)
IP Growth (MoM, Δ_r^v = 1 a.p.p.)									
Outside	-0.000	-0.001	0.005	-0.004	-0.006	0.004	-0.004	-0.005	0.004
	(0.008)	(0.006)	(0.005)	(0.009)	(0.007)	(0.006)	(0.009)	(0.007)	(0.006)
Preceding 6 Months	-0.012	0.001	-0.004	0.012	0.010	0.000	0.013	0.011	0.001
	(0.029)	(0.023)	(0.020)	(0.031)	(0.025)	(0.021)	(0.031)	(0.025)	(0.021)
During Recessions	-0.047**	-0.022	-0.019	-0.048**	-0.017	-0.018	-0.048**	-0.016	-0.016
	(0.021)	(0.017)	(0.014)	(0.022)	(0.018)	(0.015)	(0.022)	(0.018)	(0.015)
Following 6 Months	0.044	-0.007	0.000	0.044	-0.005	0.008	0.043	-0.006	0.007
	(0.041)	(0.029)	(0.023)	(0.045)	(0.032)	(0.025)	(0.045)	(0.032)	(0.025)
Real GDP Growth (QoQ, Δ_r^v = 1 a.p.p.)									
Outside	0.026	0.007	-0.027	0.019	0.003	-0.020	0.019	0.004	-0.018
	(0.055)	(0.040)	(0.033)	(0.060)	(0.043)	(0.036)	(0.060)	(0.043)	(0.036)
Preceding 6 Months	-0.101	-0.056	-0.068	0.038	0.096	0.088	0.024	0.081	0.074
	(0.169)	(0.122)	(0.103)	(0.189)	(0.138)	(0.114)	(0.189)	(0.137)	(0.114)
During Recessions	0.019	0.072	0.047	0.075	0.108	0.072	0.072	0.108	0.076
	(0.114)	(0.079)	(0.064)	(0.125)	(0.084)	(0.066)	(0.125)	(0.084)	(0.066)
Following 6 Months	0.160	0.104	0.139*	0.142	0.069	0.093	0.146	0.072	0.098
	(0.124)	(0.096)	(0.072)	(0.128)	(0.103)	(0.077)	(0.128)	(0.103)	(0.077)
FOMC Meetings/Calls (No Size)									
Outside	0.009	-0.031	0.006	0.025	-0.027	0.004	0.026	-0.027	0.004
	(0.065)	(0.052)	(0.045)	(0.070)	(0.056)	(0.048)	(0.070)	(0.056)	(0.048)
Preceding 6 Months	0.073	-0.031	-0.002	0.041	-0.146	-0.042	0.032	-0.154	-0.051
	(0.243)	(0.188)	(0.162)	(0.254)	(0.200)	(0.172)	(0.254)	(0.200)	(0.172)
During Recessions	-0.219	-0.069	-0.048	-0.300	-0.110	-0.082	-0.291	-0.100	-0.073
	(0.194)	(0.148)	(0.128)	(0.207)	(0.159)	(0.138)	(0.207)	(0.158)	(0.138)
Following 6 Months	0.149	0.242	0.161	0.289	0.432**	0.287*	0.312	0.449**	0.313*
	(0.244)	(0.186)	(0.157)	(0.269)	(0.201)	(0.170)	(0.268)	(0.201)	(0.170)
Observations	221,845	221,845	221,845	184,930	184,930	184,930	184,930	184,930	184,930
Adjusted R^2	0.087	0.087	0.087	0.120	0.120	0.120	0.120	0.120	0.120

Estimates of $\mathbb{I}_r^{ext}\Delta_r^v(\hat{\alpha}_r - \hat{\beta}_r)$ from (6): $\pi_j^e = \mathbf{x}_j'\boldsymbol{\gamma} + \sum_{r\in R}\mathbb{I}_r^{ext}\Delta_r^v(\beta_r\mathbb{I}_r^b + \alpha_r\mathbb{I}_r^a) + \varepsilon_j$

Notes: $^{***}p<0.01, ^{**}p<0.05, ^{*}p<0.10$. Robust standard errors are in parentheses. Sample is for June 1979 – May 2019. Bolded column headings are the control variables included, defined in Table 2, and the columns labeled with $w = 1, 2, 3$ denote the number of days before and after used to define the release window. Dependent variable is individuals reported on year ahead expected inflation rate from Michigan Survey of Consumers (MSC), excluding individuals with reported expected inflation rate greater than 20 percent in absolute value. For releases of the employment situation, consumer price index, industrial production, gross domestic product, and FOMC meetings/calls, the headline series used is in **bold**. Extended recessions refers to adding additional indicators for instances of releases occurring in the preceding 6 months before the offical NBER recession start month, as well as an additional indicator for the following 6 months after their end. These dates are provided in Table 10. Each row shows the estimated percentage point response of household inflation expectations to a percentage point (p.p.) or annualized percentage point (a.p.p.) change in the headline series for releases occurring within the time period denoted in *italics*. Main paper results are in Table 11.

A.3: Additional Results for Surprises

Table A.4 – Comparsion for Table 12: Size for November 1996 – May 2019

Dep Var: 1yr Inf. Exp.	**Base Controls**			**+ Sentiment and News**			**+ Macro Aggregates**		
Coef: $\Delta_r^v(\hat{\alpha}_r - \hat{\beta}_r)$	$w=1$	$w=2$	$w=3$	$w=1$	$w=2$	$w=3$	$w=1$	$w=2$	$w=3$
Unemployment Rate ($\Delta_r^v = 1$ p.p.)	-0.636	-0.578*	-0.350	-0.408	-0.318	-0.275	-0.410	-0.339	-0.292
	(0.429)	(0.319)	(0.263)	(0.453)	(0.340)	(0.279)	(0.452)	(0.339)	(0.279)
CPI Inflation (MoM, $\Delta_r^v = 1$ a.p.p.)	0.044*	0.022	0.016	0.037	0.009	0.007	0.038	0.009	0.006
	(0.025)	(0.018)	(0.015)	(0.027)	(0.019)	(0.016)	(0.026)	(0.019)	(0.016)
IP Growth (MoM, $\Delta_r^v = 1$ a.p.p.)	-0.015	-0.018**	-0.011*	-0.018*	-0.019**	-0.012*	-0.017	-0.018**	-0.011
	(0.010)	(0.008)	(0.006)	(0.011)	(0.008)	(0.007)	(0.011)	(0.008)	(0.007)
Real GDP Growth (QoQ , $\Delta_r^v = 1$ a.p.p.)	-0.024	-0.010	0.001	-0.007	0.004	0.023	-0.012	-0.000	0.024
	(0.059)	(0.048)	(0.038)	(0.063)	(0.050)	(0.040)	(0.063)	(0.050)	(0.040)
FOMC Meetings/Calls ($\Delta_r^v = 1$ p.p.)	0.031	-0.171	-0.119	0.197	-0.217	-0.079	0.208	-0.195	-0.056
	(0.494)	(0.400)	(0.343)	(0.525)	(0.416)	(0.359)	(0.521)	(0.414)	(0.356)
Observations	115,979	115,979	115,979	98,507	98,507	98,507	98,507	98,507	98,507
Adjusted R^2	0.044	0.044	0.044	0.087	0.087	0.087	0.088	0.088	0.088

Estimates of $\Delta_r^v(\hat{\alpha}_r - \hat{\beta}_r)$ from (3): $\pi_j^e = \boldsymbol{x}_j'\boldsymbol{\gamma} + \sum_{r\in R}\Delta_r^v(\beta_r \mathbb{r}_r^b + \alpha_r \mathbb{r}_r^a) + \varepsilon_j$

Notes: $^{***}p<0.01$,$^{**}p<0.05$,$^{*}p<0.10$. Robust standard errors are in parentheses. Sample is for November 1996 – May 2019. Bolded column headings are the control variables included, defined in Table 2, and the columns labeled with $w = 1, 2, 3$ denote the number of days before and after used to define the release window. Dependent variable is individuals reported on year ahead expected inflation rate from Michigan Survey of Consumers (MSC), excluding individuals with reported expected inflation rate greater than 20 percent in absolute value. For releases of the employment situation, consumer price index, industrial production, gross domestic product, and FOMC meetings/calls, the headline series used is in **bold**. Each row shows the estimated percentage point response of household inflation expectations for each percentage point (p.p.) or annualized percentage point (a.p.p.) change in the corresponding headline series.

Appendix B – MSC Questions and Summaries

This appendix provides some additional details for the MSC, including the questions that the survey records responses for that are used in the main results of the paper. To start with how the survey elicits the respondents one year-ahead expected inflation rate, the following series of questions are asked, in this order:

- **(A12):** During the next 12 months, do you think that prices in general will *go up*, or *go down*, or *stay* where they are now?
  - Responses: Go up, stay the same, go down, or don't know
- If **(A12)** =*stay the same*, MSC asks:
  - **(A12a):** Do you mean that prices will go up at the same rate as now, or that prices in general will not go up during the next 12 months?
- **(A12b):** By about what percent do you expect prices to go *(up/down)* on the average, during the next 12 months?
  - Response greater than 5%: Let me make sure that I have that correct. You said that you expect prices to go *(up/down)* by X percent. Is that correct?
  - If **(A12b)** = *Don't Know*: **(A12c):** How many cents on the dollar do you expect prices to go *(up/down)* on the average, during the next 12 months?

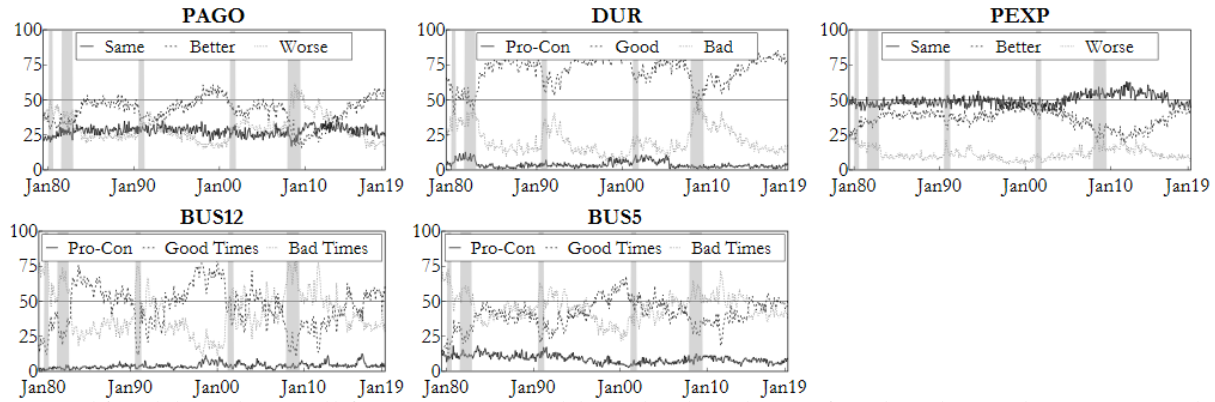
B.1: Questions Included in Sentiment Index

Table B.1 – Sentiment Questions from MSC

Label	Question	Responses
PAGO	We are interested in how people are getting along financially these days. Would you say that you (and your family living there) are better off or worse off financially than you were a year ago?	Better, same, worse
DUR	About the big things people buy for their homes – such as furniture, a refrigerator, stove, television, and things like that. Generally speaking, do you think now is a good or bad time for people to buy major household items?	Good, pro-con, bad
PEXP	Now looking ahead – do you think that a year from now you (and your family living there) will be better off financially, or worse off, or just about the same as now?	Better, same, worse
BUS12	Now turning to business conditions in the country as a whole – do you think that during the **next twelve months** we'll have good times financially, or bad times, or what?	Good times, good with qualifications, pro-con, bad with qualifications, bad times
BUS5	Looking ahead, which would you say is more likely – that in the country as a whole we'll have continuous good times during the **next five years or so**, or that we will have periods of widespread unemployment or depression, or what?	Good times, good with qualifications, pro-con, bad with qualifications, bad times

Notes: The letter abbreviations before the questions denote the label used by Michigan in their survey data, and are provided here for convenience. Additional details can be found at University of Michigan Survey of Consumers (MSC). For each question, the possible answer choices as they appear in the survey are provided in the last column. Throughout this paper, I have combined the *… with qualifications* responses to both BUS12 and BUS5 into their respective *good* and *bad* responses. The first two questions, PAGO and DUR, are the two questions used to construct the Index of Current Economic Conditions (ICC), while the last three questions are used to construct the Index of Consumer Expectations (ICE). Documentation on how these indexes are constructed can be found on the surveys website. Back to Table 2.

Figure B.1 – Responses to Sentiment Questions, June 1979 – May 2019



Notes: In each figure, the horizontal axis is monthly from June 1979 to May 2019, with the vertical axis showing the percent of respondents within a particular survey giving a particular response to one of the sentiment questions. Refer to Table B.1 for details of the questions.

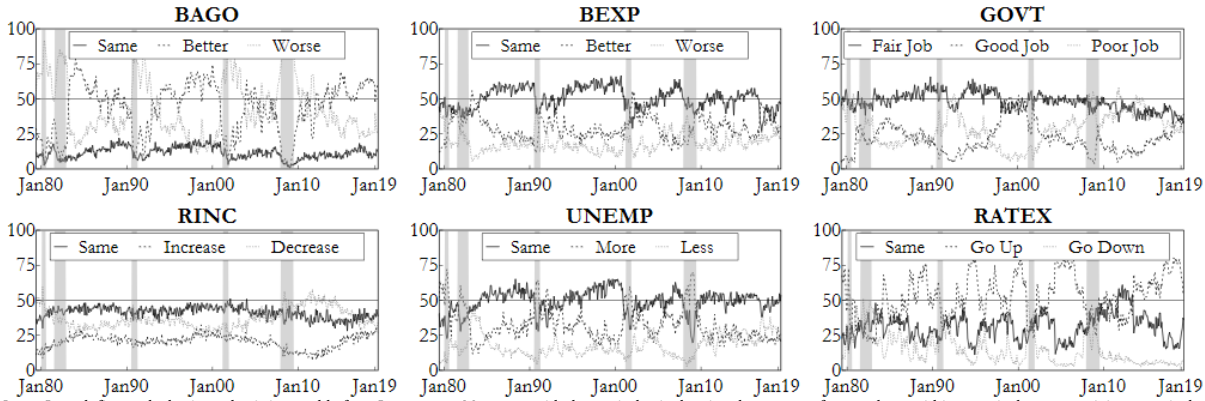
B.2: Additional Questions Included

Table B.2 – Additional Questions from MSC

Label	Question	Responses
BAGO	Would you say that at the present time business conditions are better or worse than they were a year ago?	Better, same, worse
BEXP	And how about a year from now, do you expect that in the country as a whole business conditions will be better, or worse than they are at the present, or just about the same?	Better, same, worse
GOVT	As to the economic policy of the government – I mean steps taken to fight inflation or unemployment – would you say the government is doing a good job, only fair, or a poor job?	Good, fair, poor
RINC	During the next year or two, do you expect that your (family) income will go up more than prices will go up, about the same, or less than prices will go up?	More, same, less
UNEMP	How about people out of work during the coming 12 months – do you think that there will be more unemployment than now, about the same, or less?	More, same, less
RATEX	No one can say for sure, but what do you think will happen to interest rates for borrowing money during the next 12 months – will they go up, stay the same, or go down?	Up, same, down

Notes: The letter abbreviations before the questions denote the label used by Michigan in their survey data, and are provided here for convenience. Additional details can be found at University of Michigan Survey of Consumers (MSC). For each question, the possible answer choices as they appear in the survey are provided in the last column. Back to Table 2.

Figure B.2 – Responses to Additional Survey Questions, June 1979 – May 2019



Notes: In each figure, the horizontal axis is monthly from June 1979 to May 2019, with the vertical axis showing the percent of respondents within a particular survey giving a particular response to one of the sentiment questions. Refer to Table B.2 for details of the questions. Back to Table 2.

B.3: Constructing News Heard

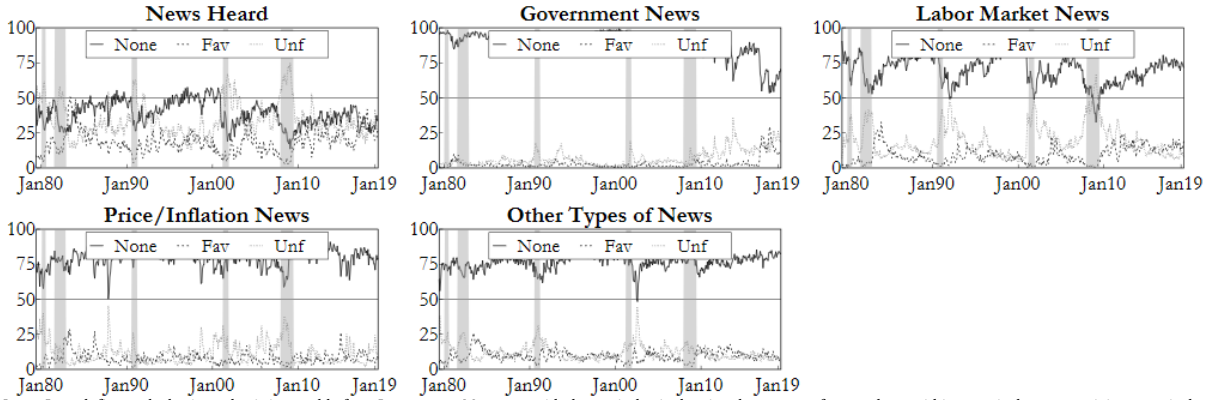
Table B.3 – News Question from MSC

News1/News2	During the last few months, have you heard of any favorable or unfavorable changes in business conditions? What did you hear?			
Category	**Code**	**Favorable Reason**	**Code**	**Unfavorable Reason**
Govt/Defense	10	Elections, admin, Congress, President	50	Election, new admin/President
	11	More military spending, more war/tensions	51	More military spending, more war/tensions
	12	Less military spending, few tensions	52	Less military spending, few tensions
	13	Gov't programs improved	53	Specific gov't spend programs changed
	14	Specific gov't programs incr/cont	54	Specific gov't spend programs eliminated
	15	Specific gov't programs decr/end	55	Gov't progams begun/increased
	16	Taxes, changes/reforms, rebates	56	Taxes, changes/reforms, rebates
	17	Other references to gov't	57	Other references to gov't
	18	Fiscal policy, budgets, deficits	58	Fiscal policy, budgets, deficits
	19	Gov't improving business condition	59	Gov't not improving business conditions
Employ/Pur. Power.	20	Opening of plants, factories, stores	60	Closing of plants, factories, stores
	21	Consumer/auto demand high	61	Consumer/auto demand low
	22	Purch power high, wages high	62	Lack of purch power, no money to spend
	23	Employ is high, plenty of jobs	63	Drop in employ, less overtime
	24	Population increase, more people to buy	64	Population increase, immigration
	25	Low debts, higher savings/assets, invest up	65	High(er) debts, lower savings/assets
	27	Other references to employ and purch power	67	Other references to employ/purch power
	28	Production increasing, GNP is up	68	Production decreasing, GNP down
	29	Unemp has risen, good for economy	69	Real estate/housing market in decline; slumping housing market
			70	Financial crisis
Prices	30	Tight money, int rates high		
	31	Lower/stable prices, less inflation	71	Prices falling, deflation
	32	Higher prices, inflation is good	72	Prices high, inflation
	33	Easier money, credit easy to get, low int rates	73	Tight money, int rates high
	34	Crowd Funding	74	Profits low, falling
	35	Profits high/rising	75	Profits high, too high
	36	Stock market, rise in price of stocks	76	Stock market decline
	37	Other references to prices/credit	77	Other price/credit references
	38	Balance of payments, dollar devalue	78	Balance of payments, dollar devalue
	39	Controls (price or wage)	79	Controls (price or wage)
Miscellaneous	40	Better race relations, less crime	80	Bad race relations; more crime
	41	Union disputes settled, relations good	81	Excessive wage demands by unions; labor unrest
	42	Times/business is good in the coming year	82	Times are bad now and won't change in next year
	43	Bad times can't last, due for good times	83	Good times can't last, due for a fall
	44	Sign of improvement already	84	Downward trends, has heard business is bad
	45	Improvements in specific industries	85	Decline in specific industries
	46	Farm situation good, crops good	86	Farm situation is bad, low farm prices, drought
	47	Other good factors or favorable ref	87	Other unfavorable/bad factors
	48	Economy more stable, optimism	88	Economy in general less stable, lack of confidence
	49	Energy crisis, pollution	89	Energy crisis, pollution, less natural resources
			90	Business/Accounting scandals

Notes: During the survey, the interviewer asks the respondent two times whether they have heard any favorable or unfavorable news. The interviewer is then responsible for classifying what category the response would fall into, including whether it would be considered favorable or unfavorable. The survey considers for broad categories of news, shown in the left most column, and within each category allows for the interviewer to record more specific reasons that the respondent cited. The columns labeled **code** denote the numbers used by the survey to classify these specific reasons into favorable or unfavorable responses. These are the codes that would appear in the variable names *news1* and *news2* that is publicly available for download from the survey website. Back to Table 2.

News Heard: The *news heard* variable I use as a control variable in Table 2 is constructed as follows. Regardless of the specific reasons recorded, I classify a respondent as having heard *favorable* news if either/both of their responses in the *news1/news2* variables are classified as being favorable, and neither of the responses would be classified as *unfavorable*. For example, a respondent may report hearing favorable news when initially prompted, which would be recorded in the *news1* variable, and then report hearing no additional news, so that *news2* would be coded as 0, denoting that the respondent did not mention any news during the second prompt. This would be classified as *favorable* news heard in my variable. An analogous method for classifying *unfavorable* news is used. In the case where respondents give conflicting responses in the *news1/news2*, for example by reporting having heard favorable price news and unfavorable government/defense news, then their news heard variable would be classified as *mixed*. Finally, respondents are classified as having heard no news if both *news1/news2* reflect such a response. Constructing the category specific news heard variables follows a similar process, with the exception being that the favorable/unfavorable classification instead requires that either of their responses do not conflict *within* the specific reasons for each category.

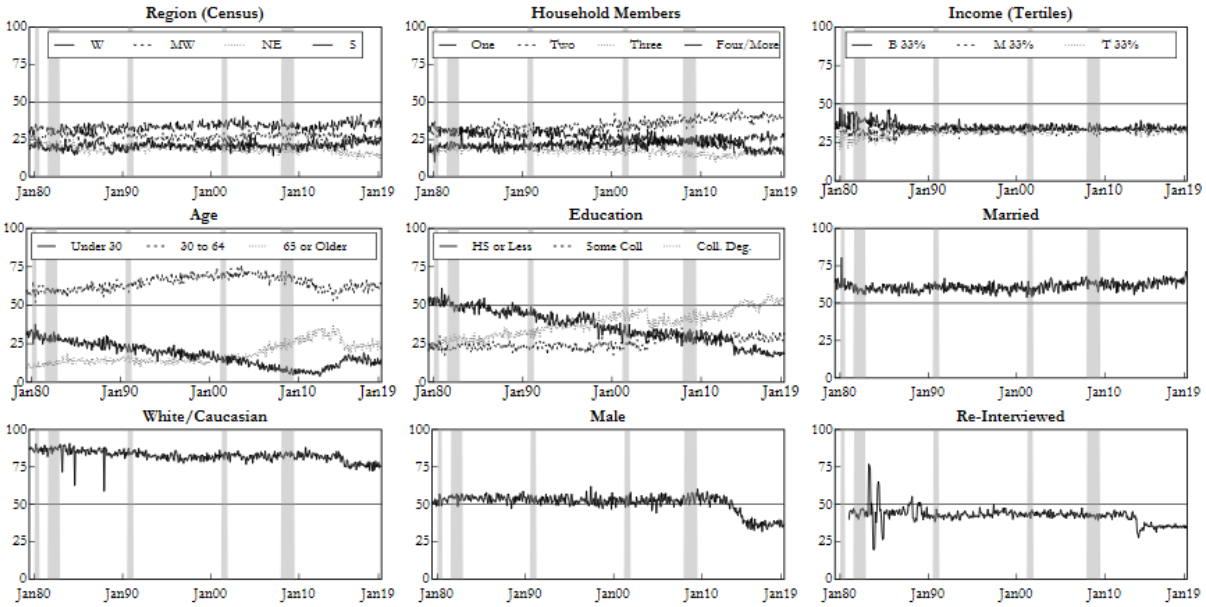
Figure B.3 – Responses to News Heard, June 1979 – May 2019



Notes: In each figure, the horizontal axis is monthly from June 1979 to May 2019, with the vertical axis showing the percent of respondents within a particular survey giving a particular response to the news heard variable. Refer to Table B.3 for additional details.

B.4: Demographics

Figure B.4 – Demographic Composition of Survey, June 1979 – May 2019



Notes: In each figure, the horizontal axis is monthly from June 1979 to May 2019, with the vertical axis showing the percent of respondents that are classified into particular groups for the demographic variables considered in Table 2.

Appendix C – Release Dates and Summaries

C.1: Historical Release Dates

Table C.1 – Release Days for Employment Situation, June 1979 – May 2019

	Release Month											
Year	**January**	**February**	**March**	**April**	**May**	**June**	**July**	**August**	**September**	**October**	**November**	**December**
1979	-	-	-	-	-	1	6	3	7	5	2	7
1980	11	1	7	4	2	6	3	1	5	3	7	5
1981	9	6	6	3	8	5	2	7	4	2	6	4
1982	8	5	5	2	7	4	2	6	3	8	5	3
1983	7	4	4	1	6	3	8	5	2	7	4	2
1984	6	3	9	6	4	1	6	3	7	5	2	7
1985	9	1	8	5	3	7	5	2	6	4	1	6
1986	8	7	7	4	2	6	3	1	5	3	7	5
1987	9	6	6	3	8	5	2	7	4	2	6	4
1988	8	5	4	1	6	3	8	5	2	7	4	2
1989	6	3	10	7	5	2	7	4	1	6	3	8
1990	5	2	9	6	4	1	6	3	7	5	2	7
1991	4	1	8	5	3	7	5	2	6	4	1	6
1992	10	7	6	3	8	5	2	7	4	2	6	4
1993	8	5	5	2	7	4	2	6	3	8	5	3
1994	7	4	4	1	6	3	8	5	2	7	4	2
1995	6	3	10	7	5	2	7	4	1	6	3	8
1996	19	2	8	5	3	7	5	2	6	4	1	6
1997	10	7	7	4	2	6	3	1	5	3	7	5
1998	9	6	6	3	8	5	2	7	4	2	22	4
1999	8	5	5	2	7	4	2	6	3	8	5	3
2000	7	4	3	7	5	2	7	4	1	6	3	8
2001	5	2	9	6	4	1	6	3	7	5	2	7
2002	4	1	8	5	3	7	5	2	6	4	1	6
2003	10	7	7	4	2	6	3	1	5	3	7	5
2004	9	6	5	2	7	4	2	6	3	8	5	3
2005	7	4	4	1	6	3	8	5	2	7	4	2
2006	6	3	10	7	5	2	7	4	1	6	3	8
2007	5	2	9	6	4	1	6	3	7	5	2	7
2008	4	1	7	4	2	6	3	1	5	3	7	5
2009	9	6	6	3	8	5	2	7	4	2	6	4
2010	8	5	5	2	7	4	2	6	3	8	5	3
2011	7	4	4	1	6	3	8	5	2	7	4	2
2012	6	3	9	6	4	1	6	3	7	5	2	7
2013	4	1	8	5	3	7	5	2	6	22	8	6
2014	10	7	7	4	2	6	3	1	5	3	7	5
2015	9	6	6	3	8	5	2	7	4	2	6	4
2016	8	5	4	1	6	3	8	5	2	7	4	2
2017	6	3	10	7	5	2	7	4	1	6	3	8
2018	5	2	9	6	4	1	6	3	7	5	2	7
2019	4	1	8	5	3	-	-	-	-	-	-	-

Notes: This table reports the release day for the U.S. Bureau of Labor Statistics (BLS) employment situation report, in which estimates of the national unemployment rate are included. To avoid potential confusion, this table shows the day a release occurred within a particular month and year, *not* the reference month of the release. These days were collected from the BLS schedules for news releases for prior years, and confirmed with Archival Economic Data (Alfred) from the Federal Reserve Bank of St. Louis.

Table C.2 – Release Days for Consumer Price Index, June 1979 – May 2019

	Release Month											
Year	**January**	**February**	**March**	**April**	**May**	**June**	**July**	**August**	**September**	**October**	**November**	**December**
1979	-	-	-	-	-	26	26	24	25	26	27	21
1980	25	22	25	22	23	24	23	22	23	24	25	23
1981	23	25	24	23	22	23	23	25	24	23	24	22
1982	22	25	23	23	21	22	23	24	23	26	23	21
1983	21	25	23	22	24	22	22	23	23	25	23	21
1984	24	24	23	24	22	22	24	22	21	24	21	20
1985	23	26	22	23	21	20	23	22	24	23	22	20
1986	22	25	25	22	21	20	23	21	23	23	25	19
1987	21	27	27	24	22	23	22	21	23	23	20	18
1988	20	26	23	20	20	21	22	23	21	21	22	20
1989	19	22	21	18	18	16	19	18	19	19	21	19
1990	18	21	20	17	16	15	18	16	18	18	16	18
1991	16	20	19	12	14	14	17	14	13	17	14	13
1992	16	19	17	10	13	12	14	13	15	15	13	11
1993	15	18	17	9	13	15	14	13	14	15	10	10
1994	13	17	16	13	13	14	13	12	13	14	16	14
1995	11	15	16	12	12	13	14	11	13	13	15	14
1996	-	1 and 28	15	12	14	12	16	13	13	16	14	12
1997	14	19	19	15	15	17	16	14	16	16	18	16
1998	13	24	19	14	14	16	14	18	17	16	17	15
1999	14	19	18	13	14	16	15	17	15	19	17	14
2000	14	18	17	14	16	14	18	16	15	18	16	15
2001	17	21	21	17	16	15	18	16	18	19	16	14
2002	16	20	21	16	15	18	19	16	18	18	19	17
2003	16	21	21	16	16	17	16	15	16	16	18	16
2004	15	20	17	14	14	15	16	17	16	19	17	17
2005	19	23	23	20	18	15	14	16	15	14	16	15
2006	18	22	16	19	17	14	19	16	15	18	16	15
2007	18	21	16	17	15	15	18	15	19	17	15	14
2008	16	20	14	16	14	13	16	14	16	16	19	16
2009	16	20	18	15	15	17	15	14	16	15	18	16
2010	15	19	18	14	19	17	16	13	17	15	17	15
2011	14	17	17	15	13	15	15	18	15	19	16	16
2012	19	17	16	13	15	14	17	15	14	16	15	14
2013	16	21	15	16	16	18	16	15	17	30	20	17
2014	16	20	18	15	15	17	22	19	17	22	20	17
2015	16	26	24	17	22	18	17	19	16	15	17	15
2016	20	19	16	14	17	16	15	16	16	18	17	15
2017	18	15	15	14	12	14	14	11	14	13	15	13
2017	18	15	15	14	12	14	14	11	14	13	15	13
2018	12	14	13	11	10	12	12	10	13	11	14	12
2019	11	13	12	10	10	-	-	-	-	-	-	-

Notes: Release days for CPI were collected from the BLS schedules for news releases for prior years, and confirmed with Archival Economic Data (Alfred) from the Federal Reserve Bank of St. Louis. The CPI release that was supposed to occur in January 1996 was delayed due to congressional gridlock over financing federal expenditures for that year. To avoid potential confusion, this table shows the day a release occurred within a particular month and year, *not* the reference month of the release.

Table C.3 – Release Days for Industrial Production, June 1979 – May 2019

	Release Month											
Year	**January**	**February**	**March**	**April**	**May**	**June**	**July**	**August**	**September**	**October**	**November**	**December**
1979	-	-	-	-	-	15	13	16	14	16	15	14
1980	16	15	14	15	16	13	16	15	16	16	14	16
1981	15	17	17	15	15	16	15	14	16	16	13	16
1982	15	17	16	15	14	15	15	13	15	15	16	15
1983	14	16	15	15	13	15	15	16	15	14	15	15
1984	13	15	15	13	15	15	13	15	14	16	15	14
1985	15	15	15	16	15	14	18	15	13	16	15	13
1986	16	14	14	15	15	16	15	15	16	16	14	15
1987	16	13	13	15	15	16	15	14	15	16	16	14
1988	15	17	16	15	16	15	15	15	14	14	15	14
1989	18	15	16	14	15	15	14	16	15	17	14	15
1990	17	16	16	17	15	15	17	15	14	17	13	14
1991	16	15	15	16	13	14	15	16	17	17	15	16
1992	17	14	17	15	15	16	15	14	16	16	16	16
1993	15	18	17	16	14	16	16	16	16	15	15	15
1994	14	15	15	15	16	15	15	15	16	14	15	14
1995	17	15	15	14	16	15	14	15	15	17	15	14
1996	24	16	15	16	15	14	16	15	17	17	15	16
1997	17	14	14	16	15	17	16	14	16	17	17	15
1998	16	17	17	17	15	16	16	14	16	16	16	16
1999	15	17	16	16	14	16	16	17	16	15	16	15
2000	14	15	15	14	15	15	14	15	15	17	15	15
2001	17	16	16	17	14	15	17	15	14	16	16	14
2002	16	15	15	16	15	14	16	15	17	17	15	17
2003	17	14	14	15	15	17	16	15	15	16	14	16
2004	16	17	15	16	14	16	15	17	15	15	17	14
2005	14	16	16	15	17	15	15	16	14	14	17	15
2006	17	15	17	14	16	15	17	16	15	17	16	15
2007	17	15	16	17	16	15	17	15	14	16	16	14
2008	16	15	17	16	15	17	16	15	15	16	17	15
2009	16	18	16	15	15	16	15	14	16	16	17	15
2010	15	17	15	15	14	16	15	17	15	18	16	15
2011	14	16	17	15	17	15	15	16	15	17	16	15
2012	18	15	16	17	16	15	17	15	14	16	16	14
2013	16	15	15	16	15	14	16	15	16	28	15	16
2014	17	14	17	16	15	16	16	15	15	16	17	15
2015	16	18	16	15	15	15	15	14	15	16	17	16
2016	15	17	16	15	17	15	15	16	15	17	16	14
2017	18	15	17	18	16	15	14	17	15	17	16	15
2018	17	15	16	17	16	15	17	15	14	16	16	14
2019	18	15	15	16	15	-	-	-	-	-	-	-

Notes: Release days for Industrial Production were collected from Board of Governors. To avoid potential confusion, this table shows the day a release occurred within a particular month and year, *not* the reference month of the release.

Table C.4 – Release Days for Gross Domestic Product, June 1979 – May 2019

	Release Month											
Year	**January**	**February**	**March**	**April**	**May**	**June**	**July**	**August**	**September**	**October**	**November**	**December**
1979	-	-	-	-	-	20	20	17	19	16	20	19
1980	18	22	19	18	20	18	18	19	19	17	19	23
1981	21	19	18	20	19	18	22	19	18	20	19	18
1982	20	22	19	21	19	21	21	19	21	20	19	21
1983	19	22	21	20	19	21	21	19	21	20	22	21
1984	20	17	20	19	18	20	23	20	20	19	20	19
1985	22	21	21	18	21	20	18	20	20	17	20	20
1986	22	20	19	17	20	18	22	19	18	22	19	17
1987	22	19	18	23	22	17	24	21	18	23	24	17
1988	27	25	23	26	26	23	27	25	20	26	29	20
1989	27	28	23	26	25	22	27	29	21	26	29	20
1990	26	28	28	27	24	21	27	24	25	30	28	19
1991	25	27	27	26	29	26	26	28	26	29	-	4 and 20
1992	29	28	26	28	29	25	30	27	24	27	25	22
1993	28	26	26	29	28	23	29	31	29	28	-	1 and 22
1994	28	-	1 and 31	28	27	29	29	26	29	28	30	22
1995	27	-	1 and 31	28	31	30	28	30	29	27	-	1 and 22
1996	19	23	-	2	2 and 30	28	-	1 and 29	27	30	27	20
1997	31	28	28	30	30	27	31	28	26	31	26	23
1998	30	27	26	30	28	25	31	27	24	30	24	23
1999	29	26	31	30	27	25	29	26	30	28	24	22
2000	28	25	30	27	25	29	28	25	28	27	29	21
2001	31	28	29	27	25	29	27	29	28	31	30	21
2002	30	28	28	26	24	27	31	29	27	31	26	20
2003	30	28	27	25	29	26	31	28	26	30	25	23
2004	30	27	25	29	27	25	30	27	29	29	30	22
2005	28	25	30	28	26	29	29	31	29	28	30	21
2006	27	28	30	28	25	29	28	30	28	27	29	21
2007	31	28	29	27	31	28	27	30	27	31	29	20
2008	30	28	27	30	29	26	31	28	26	30	25	23
2009	30	27	26	29	29	25	31	27	30	29	24	22
2010	29	26	26	30	27	25	30	27	30	29	23	22
2011	28	25	25	28	26	24	29	26	29	27	22	22
2012	27	29	29	27	31	28	27	29	27	26	29	20
2013	30	28	28	26	30	26	31	29	26	-	7	5 and 20
2014	30	28	27	30	29	25	30	28	26	30	25	23
2015	30	27	27	29	29	24	30	27	25	29	24	22
2016	29	26	25	28	27	28	29	26	29	28	29	22
2017	27	28	30	28	26	29	28	30	28	27	29	21
2018	26	28	28	27	30	28	27	29	27	26	28	21
2019	-	28	28	26	30	-	-	-	-	-	-	-

Notes: This table reports the release day for the U.S. Bureau of Economic Analysis (BEA) Gross Domestic Product report. To avoid potential confusion, this table shows the day a release occurred within a particular month and year, without regard to the revision number or reference month. Historically, the first month of a quarter includes the advanced estimate of GDP, with revisions in the following two months of the same quarter. These values would be estimates for the preceeding quarter. These days were collected from Archival Economic Data (Alfred) from the Federal Reserve Bank of St. Louis.

Table C.5 – FOMC Meeting and Call Days, June 1979 – May 2019

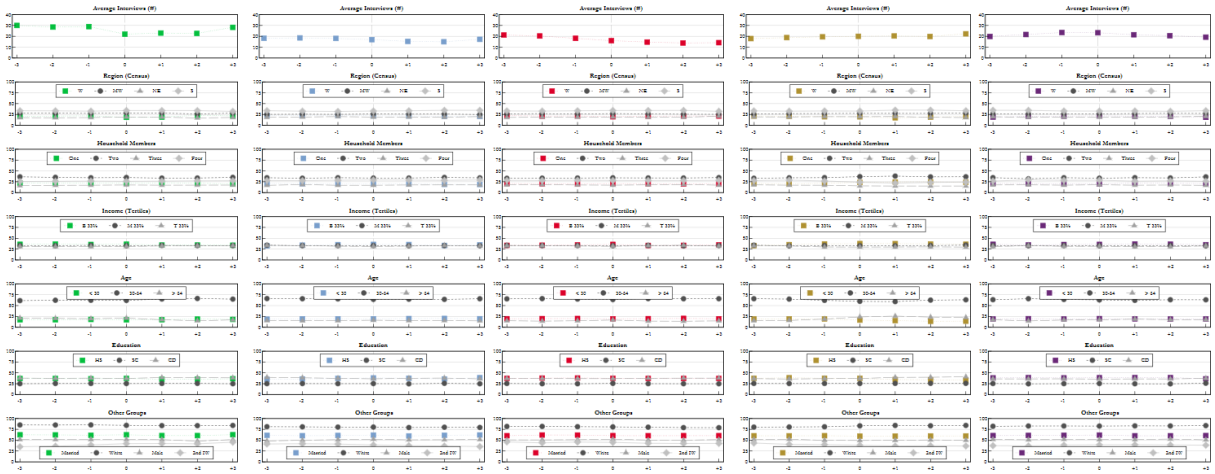
	Calendar Month											
Year	**January**	**February**	**March**	**April**	**May**	**June**	**July**	**August**	**September**	**October**	**November**	**December**
1979	-	-	-	-	-	27^{c}	11	14	18	5^{c}	20	-
1980	8*	4*	7^{c}	22	6^{c}	5^{c}	9	12	16	21	18	5^{c}
1981	-	2*	31	28^{c}	6^{c}	17^{c}	6*	7^{c}	-	5*	17	21*
1982	-	1*	29*	-	18	30*	15^{c}	24	24^{c}	5	16	20*
1983	14^{c}	8*	28*	29^{c}	24	23^{c}	12*	23	8^{c}	4	14*	19*
1984	30*	-	20^{c}	-	21*	-	16*	21	-	2	7	17*
1985	18^{c}	12*	26	-	21	-	9*	20	23^{c}	1	4*	16*
1986	-	11*	-	1	20	-	8*	19	23	-	5	15*
1987	-	10*	31	29^{c}	19	-	7	18	22	19^{c}	3	15*
1988	5^{c}	9*	29	-	17	29*	19^{c}	16	20	17^{c}	1	13*
1989	-	7*	28	-	16	5^{c}	5*	22	-	3	14	18*
1990	-	6*	27	11^{c}	15	-	2*	21	7^{c}	2	13	7^{c}
1991	9^{c}	1^{c}	26	12^{c}	14	10^{c}	2*	5^{c}	13^{c}	1	5	2^{c}
1992	9^{c}	4*	11^{c}	-	19	30*	-	18	-	6	17	14^{c}
1993	6^{c}	2*	1^{c}	-	18	-	6*	17	21	5^{c}	9^{c}	21
1994	-	3*	22	18^{c}	17	-	5*	16	27	-	15	20
1995	13^{c}	-	10^{c}	28^{c}	23	-	5*	22	26	-	15	19
1996	30*	-	26	-	21	-	2*	20	24	-	13	17
1997	-	4*	25	-	20	-	1*	19	30	-	12	16
1998	-	3*	31	-	19	30*	-	18	21^{c}	15^{c}	17	22
1999	-	2*	30	-	18	29*	-	24	-	5	16	21
2000	-	1*	21	-	16	27*	-	22	-	3	15	19
2001	3^{c}	-	20	11^{c}	15	26*	-	21	13^{c}	2	6	11
2002	29*	-	19	-	7	25*	-	13	24	-	6	10
2003	28*	-	18	1^{c}	6	24*	-	12	15*	28	-	9
2004	27*	-	16	-	4	29*	-	10	21	-	10	14
2005	-	1*	22	-	3	29*	-	9	20	-	1	13
2006	31	-	27*	-	10	28*	-	8	20	24*	-	12
2007	30*	-	20*	-	9	27*	-	7	18	30*	-	6
2008	9	-	10	29*	-	24*	24	5	16	7	-	15*
2009	16	7	17*	28*	-	3	-	11*	22*	-	3*	15*
2010	26*	-	16	27*	9	22*	-	10	21	15	2*	14
2011	25*	-	15	26*	-	21*	-	1	20*	-	1*	13
2012	24*	-	13	24*	-	19*	31*	-	12*	23*	-	11*
2013	29*	-	19*	30*	-	18*	30*	-	17*	16	-	17*
2014	28*	-	4	29*	-	17*	29*	-	16*	28*	-	16*
2015	27*	-	17*	28*	-	16*	28*	-	16*	-	-	15*
2016	26*	-	15*	26*	-	14*	26*	-	20*	-	1*	13*
2017	31*	-	14*	-	2*	13*	25*	-	19*	31*	-	12*
2018	30*	-	20*	-	1*	12*	31*	-	25*	-	7*	18*
2019	29*	-	19*	30*	-	18*	30*	-	17*	4	-	10*

Notes: Meeting dates come from Nakamura and Steinsson (2018) and the Board of Governors website. The days marked by $*$ denote two day meetings, where the day shown is the *first* day of the meeting. The days marked with the superscript c denote conference calls.

C.2: Release Summaries

Figure C.1 – Demographics Composition of Respondents Around Releases

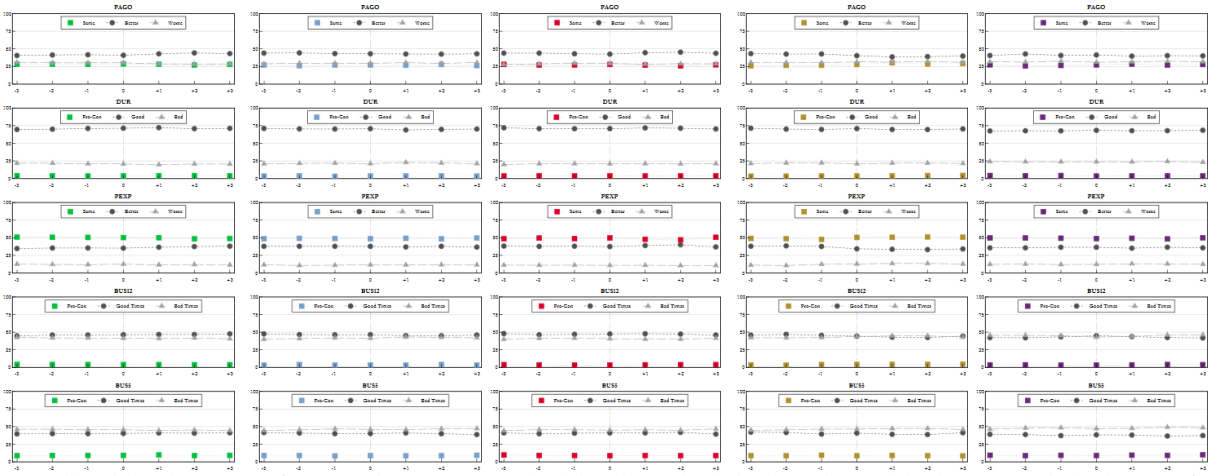
(a) EMP (b) CPI (c) IP (d) GDP (e) FOMC



Notes: Each panel shows the composition of some selected demographic variables within the three day window around the release types considered in the main text for June 1979 – May 2019. In each figure, the horizontal axis denotes the number of days before/after the release type, where the negatives refer to interviews conducted before and 0 denotes the day of release. The vertical axis in each figure is the percent of respondents falling into one of the particular categories shown. Back to Table 2.

Figure C.2 – Sentiment Responses Around Releases

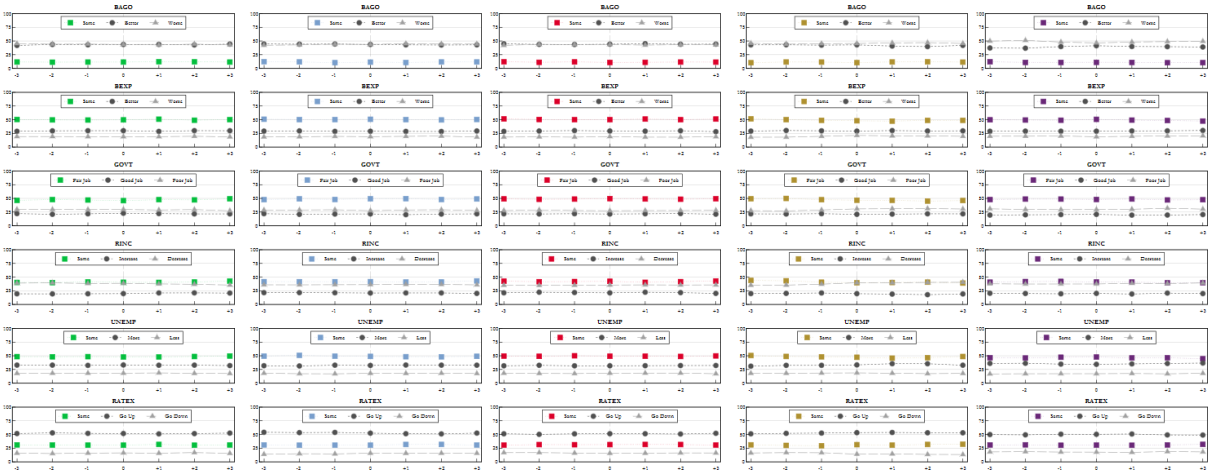
(a) EMP (b) CPI (c) IP (d) GDP (e) FOMC



Notes: Each panel shows the composition of responses to the sentiment questions detailed in Table B.1 within the three day window around the release types considered in the main text for June 1979 – May 2019. In each figure, the horizontal axis denotes the number of days before/after the release type, where the negatives refer to interviews conducted before and 0 denotes the day of release. The vertical axis in each figure is the percent of respondents answering the question with a particular response. Back to Table 2.

Figure C.3 – Additional Question Responses Around Releases

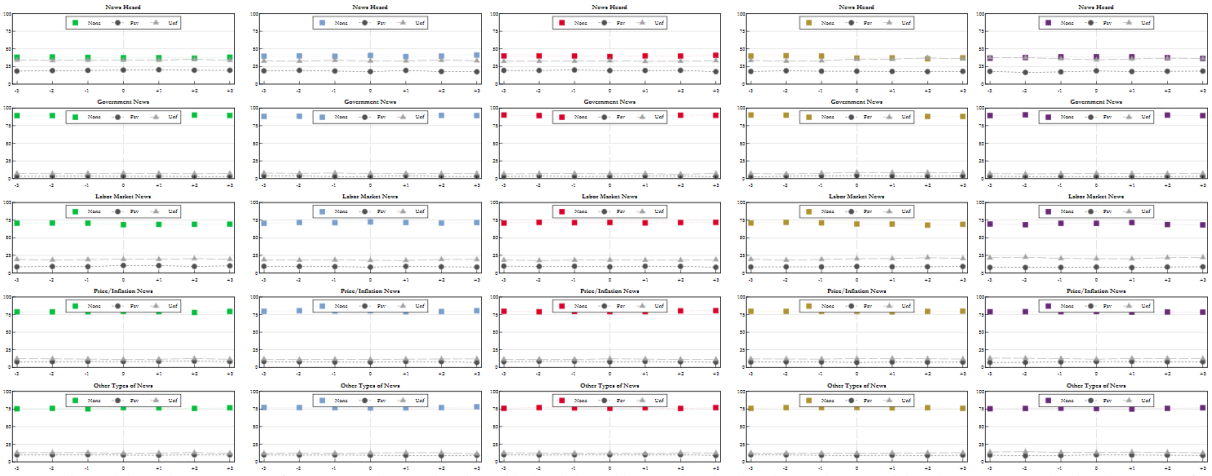
(a) EMP (b) CPI (c) IP (d) GDP (e) FOMC



Notes: Each panel shows the composition of responses to the additional questions detailed in Table B.2 within the three day window around the release types considered in the main text for June 1979 – May 2019. In each figure, the horizontal axis denotes the number of days before/after the release type, where the negatives refer to interviews conducted before and 0 denotes the day of release. The vertical axis in each figure is the percent of respondents answering the question with a particular response. Back to Table 2.

Figure C.4 – News Heard Around Releases

(a) EMP (b) CPI (c) IP (d) GDP (e) FOMC



Notes: Each panel shows the composition of responses to the news heard variable detailed in Table B.3 within the three day window around the release types considered in the main text for June 1979 – May 2019. In each figure, the horizontal axis denotes the number of days before/after the release type, where the negatives refer to interviews conducted before and 0 denotes the day of release. The vertical axis in each figure is the percent of respondents answering the question with a particular response. Back to Table 2.